

Prerequisites for LTI Systems (Revision of Linearity & Time Variance)

- » System linearity is independent of time scaling
- » System linearity is independent of coefficient used in system relationship
- » If any added/subtracted term other than i/p and o/p is available in the system relationship then the system will be nonlinear
- » If output is summation of time shifted terms of input, then the system will be linear
- » Integral and Differential operators are linear operators
- » Even and Odd operators are linear operators
- » Real, Imaginary and Conjugate operators are nonlinear operators
- » Trigonometric, Inverse Trigonometric, Logarithmic, Exponential, Roots, Powers, Modulus, sgn, sa, sinc,.....
- » For zero i/p, o/p is also equal to zero
- » Split systems are linear systems

For TIV System:

- » No time scaling
- » Coefficient in system relationship should be constant
- » Any added/subtracted term in the system relationship (except i/p and o/p) must be constant or zero

signal over a common channel with successful detection.

Orthogonal Signals: Two signals are said to be orthogonal if they are mutually independent.

Signal space $\rightarrow$ inner prod
(D.I.)

$x_1(t)$ & $x_2(t)$ N.P.S.

$$\int_{-\infty}^{\infty} x_1(t) x_2(t) dt = 0$$

$$|\vec{a}_b| \cdot |\vec{b}| \neq 0$$

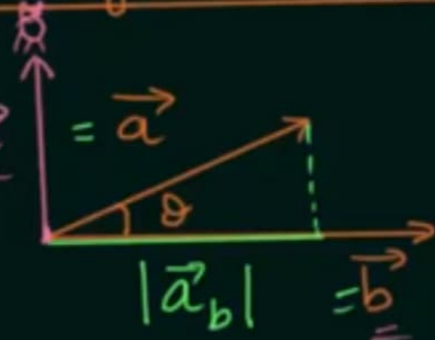
$$|\vec{a}| \cos \theta \cdot |\vec{b}| \neq 0$$

$$|\vec{a}| |\vec{b}| \cos \theta \neq 0$$

$$\vec{a} \cdot \vec{b} \neq 0$$

Dot prod / scalar prod

Orthogonal Vectors:



$$\theta = 90^\circ$$

$$\vec{a} \cdot \vec{b} = 0$$

$$\int x_1(t) x_2(t) dt = 0$$

$$\bigoplus_{-\infty}^{\infty}$$

↙ P.S.

$$\int_0^{\infty} x_1(t) x_2(t) dt = 0$$

$$\vec{a} \cdot \vec{b} \neq 0$$

Dot prod / scalar prod

$$\vec{c} \cdot \vec{b} = 0$$

Properties: i) Two harmonics of diff. f are always 0

$$x_1(t) = \sin(n\omega t + \phi_1)$$

$$x_2(t) = \sin(m\omega t + \phi_2)$$

$$n \neq m$$

$$\int_0^T \sin(n\omega t + \phi_1) \sin(m\omega t + \phi_2) dt$$

$$\int_{-\infty}^{\infty} x_1(t) x_2(t) dt = 0$$

$\oplus$

P.S.

$$\int_0^{\infty} x_1(t) x_2(t) dt = 0$$

$$|a||b|\cos\theta \neq 0$$

$$\vec{a} \cdot \vec{b} \neq 0$$

Dot prod / scalar prod

$$\vec{c} \cdot \vec{b} = 0$$

Properties: i) Two harmonics of diff. f are always 0

$$x_1(t) = \sin(n\omega_0 t + \phi_1)$$

$$x_2(t) = \sin(m\omega_0 t + \phi_2)$$

$n \neq m$

$$\int_0^T \sin(n\omega_0 t + \phi_1) \sin(m\omega_0 t + \phi_2) dt = 0$$

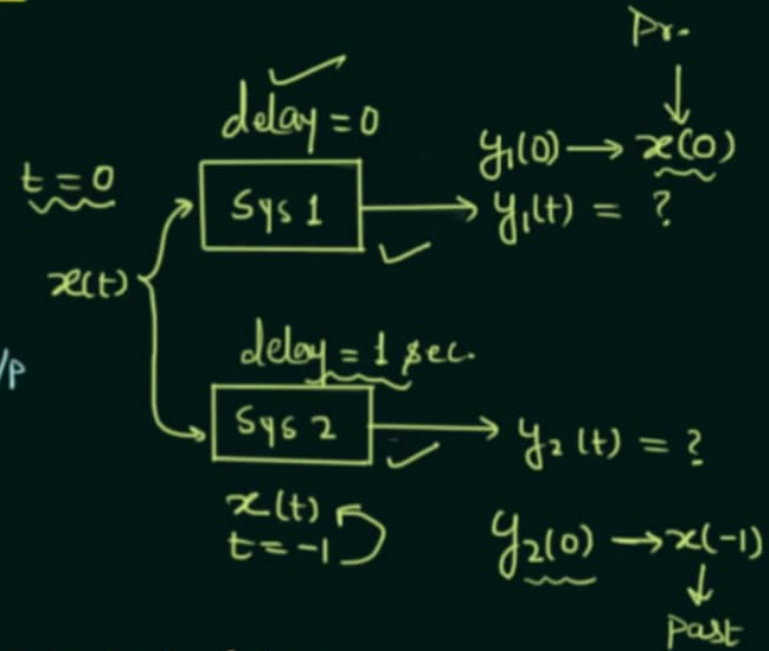
Causal and Non-Causal Systems

Causal System: o/p of system is independent of future values of i/p

ex: - i) $\check{y}(t) = \check{x}(t)$ C

ii) $\check{y}(t) = \underbrace{\check{x}(t)}_{\text{pr.}} + \underbrace{\check{x}(t-1)}_{\text{past}}$ C

C.S. $\rightarrow$ o/p $\rightarrow$ P i/p +
o/p $\rightarrow$ past i/p



Non-Causal System: o/p of system depends on future values of i/p at any instant of time

ex: - i) $\check{y}(t) = \underbrace{\check{x}(t+2)}_{\text{fut.}}$ N.C. + A.C.

ii) $\check{y}(t) = \underbrace{\check{x}(t)}_{\text{pr.}} + \underbrace{\check{x}(t-1)}_{\text{past}} + \underbrace{\check{x}(t+1)}_{\text{fut.}}$ N.C.

Anti-Causal System:

o/p $\rightarrow$ only on fut values of i/p

Time-Invariant & Time-Variant Systems (Solved Problems)

For TIV Systems:

1. No time scaling
2. Coeff. should be constant
3. Any added/subtracted term in the system relationship (except i/p and o/p) must be constant or zero

10. $y(t) = 1 \cdot x(t+1) + 1 \cdot x(t-1)$

1. ✓

2. ✓

3. ✓

TIV Sys.

11. $y(t) = \int_{-\infty}^t x(z) dz$

TIV Sys.

12. $y(t) = \int_{-\infty}^t x(3z) dz$

TV Sys.

Time-Invariant & Time-Variant Systems (Solved Problems)

For TIV Systems:

1. No time scaling
2. Coeff. should be constant
3. Any added/subtracted term in the system relationship (except i/p and o/p) must be constant or zero

10. $y(t) = 1 \cdot x(t+1) + 1 \cdot x(t-1)$ 11. $y(t) = \int_{-\infty}^t x(\tau) d\tau$ 12. $y(t) = \int_{-\infty}^t x(3\tau) d\tau$

1. ✓ TIV Sys. 2. ✓ 3. ✓

13. $y(t) = \int_{-\infty}^t \cos \tau \cdot x(\tau) d\tau$ TIV Sys. TV Sys.

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Time-Invariant and Time-Var...
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Time-Invariant and
Time-Variant Systems (Split ...
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Linear and Non-Linear
Systems
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Linear and Non-Linear
Systems (Solved Problems) | ...
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Linear & Non-Linear Systems (Solved Problems) PART 2

Linear and Non-Linear
Systems (Solved Problems) | ...
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Linear and Non-Linear Systems (Solved Problems)

>> System linearity is independent of time scaling

system $\rightarrow$ time scaling
+
amp. shifting
+
coeff. etc.

$$4. \quad y(t) = \underbrace{\sin t}_{\text{coeff}} \cdot \underbrace{x(t)}_{\text{i/p}}$$

$$x(t) \rightarrow \boxed{\text{System}} \rightarrow y(t) = \sin t \cdot x(t)$$

a) L.O.A

$$x_1(t) \rightarrow \text{sys.} \rightarrow y_1(t) = \sin t \cdot x_1(t)$$

$$x_2(t) \rightarrow \text{sys.} \rightarrow y_2(t) = \sin t \cdot x_2(t)$$

$$y_1(t) + y_2(t) = \sin t [x_1(t) + x_2(t)]$$

$$\underline{x_1(t) + x_2(t)} \rightarrow \text{sys.} \rightarrow \sin t \cdot \underline{x_1(t) + x_2(t)}$$

b) L.O.H

$$x(t) \rightarrow \text{sys.} \rightarrow y(t) \rightarrow 'k' \rightarrow ky(t) = k \sin t \cdot x(t)$$

$$kx(t) \rightarrow \text{sys.} \rightarrow \sin t \cdot kx(t) = k \sin t \cdot x(t)$$

$\therefore$ follows P.O.S
Linear Sys.

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1.5K

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Both are linear system

Linear & Non-Linear Systems (Solved Problems)

PART 3

>> System linearity is independent of time scaling

>> System linearity is independent of coefficient used in system relationship

$$6. \quad y(t) = \underbrace{2t}_{\text{o/p}} + \underbrace{x(t)}_{\text{i/p}}$$

$$x(t) \rightarrow \text{sys.} \rightarrow y(t) = 2t + x(t)$$

a) L.O.A

$$x_1(t) \rightarrow \text{sys.} \rightarrow y_1(t) = 2t + x_1(t)$$

$$7. \quad y(t) = \underbrace{-2}_{\text{o/p}} + \underbrace{t}_{\text{time ind}} + \underbrace{x(t)}_{\text{i/p}}$$

$$x(t) \rightarrow \text{sys.} \rightarrow y(t) = -2 + t + x(t)$$

a) L.O.H

$$x(t) \rightarrow \text{sys.} \rightarrow y(t) \rightarrow 'k' \rightarrow ky(t)$$

$$kx(t) \rightarrow \text{sys.} \rightarrow -2 + t + kx(t)$$

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Signal & System

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Linear and Non-Linear Systems (Solved Problems) | Part 3

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Time Domain: variable $\rightarrow$ real

LTI Sys.

impulse response
i/p o/p

$$\underline{y(t)} = x(t-1) + x(t+1)$$

Freq. Domain:

LT
 $\downarrow$

$$Y(s) = \underline{X(s)} e^{-s} + \underline{X(s)} e^s$$

$$\underline{H(s)} = \frac{Y(s)}{X(s)} \Big|_{\text{all the initial cond}^n = 0}$$

$$Y(s) = X(s) (e^{-s} + e^s)$$

$$\underline{H(s)} = e^{-s} + e^s$$

$f(t)$

$\downarrow$ LT

$$L\{f(t)\} = F(s) = \int_{-\infty}^{\infty} \underline{f(t)} \cdot e^{-st} dt$$

complex complex

$$\underline{s} = \sigma + j\omega$$

$$\int_{-\infty}^0 0 = 0$$

$f(t) = 2 \cdot u(t)$

$$F(s) = \int_{-\infty}^{\infty} 2 \cdot e^{-st} dt$$

$$= 2 \int_{-\infty}^{\infty} e^{-st} dt$$

$$= 2 \cdot \frac{1}{-s} [e^{-st}]_{-\infty}^{\infty}$$

$$\underline{y(t)} = x(t-1) + x(t+1)$$

Freq. Domain: $\begin{matrix} \uparrow \text{ILT} \\ \downarrow \text{LT} \end{matrix}$

$$Y(s) = \underline{X(s)} e^{-s} + \underline{X(s)} e^s$$

$$h(t) \rightleftharpoons H(s)$$

$$\underline{H(s)} = \frac{Y(s)}{X(s)} \Big|_{\text{all the initial cond}^n = 0}$$

$$Y(s) = X(s)(e^{-s} + e^s)$$

$$\underline{H(s)} = \underline{e^{-s}} + e^s$$

$\begin{matrix} \uparrow \text{ILT} & \downarrow \text{ILT} \\ \text{LT} & \text{ILT} \end{matrix}$

$$\underline{h(t)} = \underline{\delta(t-1)} + \delta(t+1)$$

$$f(t) \xrightarrow{\text{LT}}$$

$$L\{f(t)\} = F(s) = \int_{-\infty}^{\infty} \underbrace{f(t)}_{\text{complex}} \cdot \underbrace{e^{-st}}_{\text{complex}} dt$$

$t < 0$
 $t \geq 0$

$$s = \sigma + j\omega$$

$$\int_{-\infty}^0 0 = 0$$

$$\underline{f(t)} = 2 \cdot \underline{u(t)}$$

$$F(s) = \int_{-\infty}^{\infty} 2 \cdot e^{-st} dt$$

$$= 2 \int_0^{\infty} e^{-st} dt$$

$$= 2 \cdot \frac{1}{s} [e^{-st}]_0^{\infty}$$

$$= -\frac{2}{s} [e^{-\infty} - e^0]$$

$$\underline{f(t)} = \underline{\delta(t-2)}$$

$$\downarrow \text{LT}$$

$$F(s) = e^{-2s}$$

$$L\{2u(t)\} = 2/s$$

$$L\{au(t)\} = a/s$$

Transfer Function Calculation

non LTI



$$1. y(t) = x(t-1) + 5$$

$$H(s) = ?$$

$$= \frac{Y(s)}{X(s)}$$

LT

$$Y(s) = X(s) \cdot e^{-s} + \infty$$

$$\left. \begin{aligned} L\{2\} &= \infty \\ L\{5\} &= \infty \\ L\{dc\} &= \infty \end{aligned} \right\} \text{undefined}$$

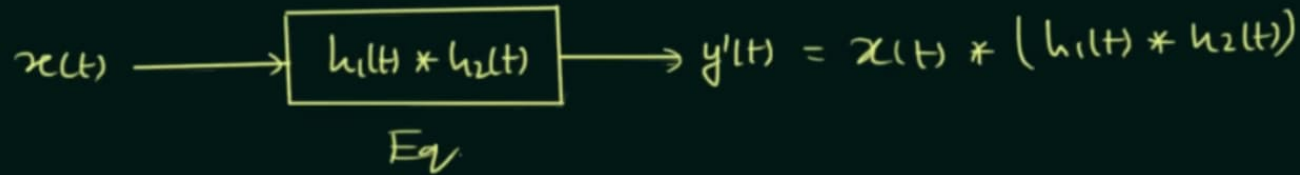
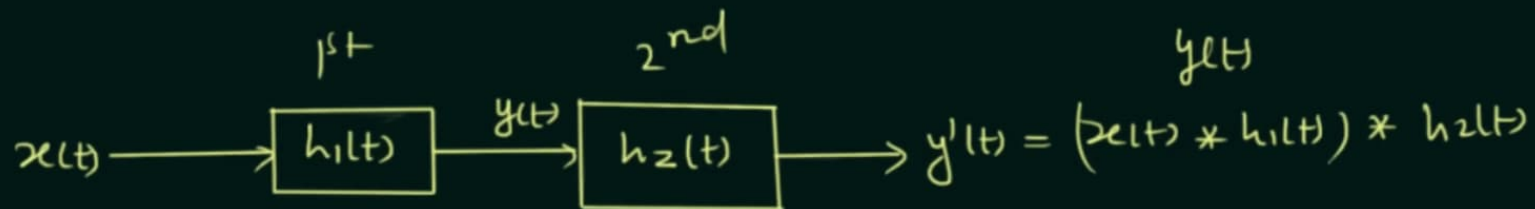
system is known LTI because we have 5
here 5 is the added Tom and whenever

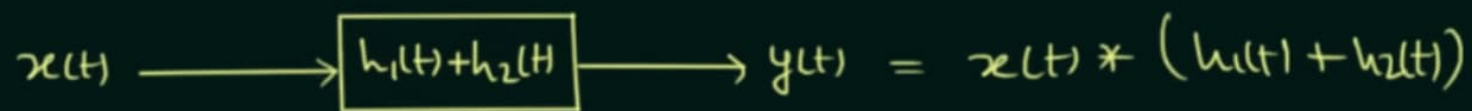
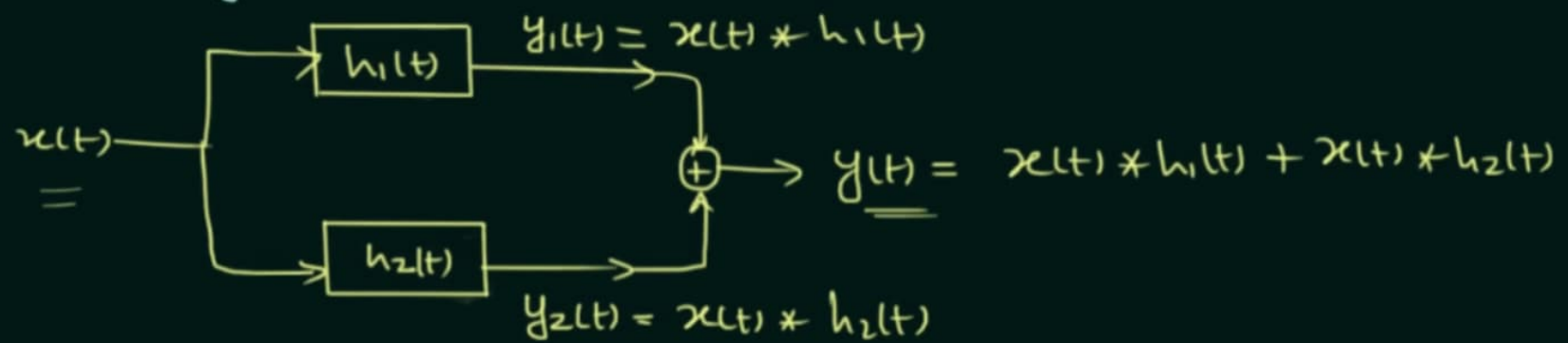
$$Y(s) = X(s)e^{-s} + 5/s$$

i/p i/m. res. i/p i/m. res.

2. Associative Property:

$$(x_1(t) * x_2(t)) * x_3(t) = y(t) = x_1(t) * (x_2(t) * x_3(t))$$





Properties of Fourier Transform (Part 1)

1. Linearity:

$$x_1(t) \iff X_1(j\omega)$$

$$x_2(t) \iff X_2(j\omega)$$

$$\alpha x_1(t) \iff \alpha X_1(j\omega)$$

$$\beta x_2(t) \iff \beta X_2(j\omega)$$

$$\checkmark \alpha x_1(t) + \beta x_2(t) \iff \alpha X_1(j\omega) + \beta X_2(j\omega)$$

proof:- $x(t) \iff X(j\omega)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

and in this way we have proved the conjugation property so this is all for

$$\begin{aligned} &= \int_{-\infty}^{\infty} [\alpha x_1(t) + \beta x_2(t)] \cdot e^{-j\omega t} dt \\ &= \alpha \int_{-\infty}^{\infty} x_1(t) \cdot e^{-j\omega t} dt + \beta \int_{-\infty}^{\infty} x_2(t) \cdot e^{-j\omega t} dt \end{aligned}$$

2. Conjugation:

$$x(t) \iff X(j\omega)$$

$$\checkmark \checkmark x^*(t) \iff X^*(-j\omega)$$

proof:- $x(t) \iff X(j\omega)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$X^*(j\omega) = \int_{-\infty}^{\infty} x^*(t) \cdot e^{j\omega t} dt$$

$$\xrightarrow{\omega \rightarrow -\omega} X^*(-j\omega) = \int_{-\infty}^{\infty} x^*(t) \cdot e^{-j\omega t} dt$$

$$x^*(t) \iff X^*(-j\omega)$$

Properties of Fourier Transform (Part 1)

1. Linearity:

$$x_1(t) \Rightarrow X_1(j\omega)$$

$$x_2(t) \Rightarrow X_2(j\omega)$$

$$\alpha x_1(t) \Rightarrow \alpha X_1(j\omega)$$

$$\beta x_2(t) \Rightarrow \beta X_2(j\omega)$$

$$\alpha x_1(t) + \beta x_2(t) \Rightarrow \alpha X_1(j\omega) + \beta X_2(j\omega)$$

proof:-

$$x(t) \Rightarrow X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} [\alpha x_1(t) + \beta x_2(t)] \cdot e^{-j\omega t} dt$$

$$= \alpha \int_{-\infty}^{\infty} x_1(t) \cdot e^{-j\omega t} dt + \beta \int_{-\infty}^{\infty} x_2(t) \cdot e^{-j\omega t} dt$$

2. Conjugation:

$$x(t) \Rightarrow X(j\omega)$$

$$x^*(t) \Rightarrow X^*(-j\omega)$$

proof:-

$$x(t) \Rightarrow X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$X^*(j\omega) = \int_{-\infty}^{\infty} x^*(t) \cdot e^{j\omega t} dt$$

$$X^*(-j\omega) = \int_{-\infty}^{\infty} x^*(t) \cdot e^{-j\omega t} dt$$

$$x^*(t) \Rightarrow X^*(-j\omega)$$

Properties of Fourier Transform (Part 6)

10. Multiplication in Time: (Convolution in Frequency)

$$x_1(t) \Rightarrow X_1(j\omega)$$

$$x_2(t) \Rightarrow X_2(j\omega)$$

$$x_1(t) \cdot x_2(t) \Rightarrow \frac{1}{2\pi} \{X_1(j\omega) * X_2(j\omega)\}$$

$$\text{Proof: } - \text{FT} \{x_1(t) \cdot x_2(t)\} = \frac{1}{2\pi} \{X_1(j\omega) * X_2(j\omega)\}$$

$$\text{or } x_1(t) \cdot x_2(t) = \text{IFT} \left[\frac{1}{2\pi} \{X_1(j\omega) * X_2(j\omega)\} \right]$$

$$x(t) = \text{IFT} \{X(j\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega$$

$$\text{IFT} \left\{ \frac{1}{2\pi} \{X_1(j\omega) * X_2(j\omega)\} \right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(j\omega) * X_2(j\omega) \right\} \cdot e^{j\omega t} d\omega$$

$$= \left(\frac{1}{2\pi} \right)^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X_1(j\omega) \cdot X_2(j(\omega - \lambda)) d\lambda \cdot e^{j\omega t} d\omega$$

$$= \left(\frac{1}{2\pi} \right)^2 \int_{-\infty}^{\infty} X_1(j\lambda) \cdot e^{j\lambda t} d\lambda \int_{-\infty}^{\infty} X_2(j(\omega - \lambda)) \cdot e^{j(\omega - \lambda)t} d\omega$$

$$\omega - \lambda = \beta \Rightarrow d\omega = d\beta$$

$$\omega \rightarrow -\infty \Rightarrow \beta \rightarrow -\infty$$

$$\omega \rightarrow \infty \Rightarrow \beta \rightarrow \infty$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega$$

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181 Signal & System Fourier Transform Properties PART 6

Properties of Fourier Transfo...
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182 Signal & System Fourier Transform Properties PART 7

Properties of Fourier Transform (Part 7)
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184 Signal & System Parseval's Energy Theorem

Parseval's Energy Theorem
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Duality Property of Fourier Transform

Duality Property of Fourier Transform
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Properties of Fourier Transform (Part 8)

Differentiation in Frequency:

$$x(t) \Leftrightarrow X(j\omega)$$

$$t \cdot x(t) \Leftrightarrow (j) \frac{d}{d\omega} X(j\omega) \quad \eta = 1$$

Proof: -

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot \frac{d}{d\omega} e^{-j\omega t} dt$$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot (-j) \cdot t \cdot e^{-j\omega t} dt$$

$$j \frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} t \cdot x(t) \cdot e^{-j\omega t} dt$$

FT $\xleftarrow{\text{signal}}$

14. Modulation: $x(t) \Leftrightarrow X(j\omega)$ or $x(\omega)$

i) $x(t) \cdot \cos \omega_0 t \Leftrightarrow \frac{1}{2} [X(j(\omega + \omega_0)) + X(j(\omega - \omega_0))]$

ii) $x(t) \cdot \sin \omega_0 t \Leftrightarrow \frac{j}{2} [X(j(\omega + \omega_0)) - X(j(\omega - \omega_0))]$

Proof: - i) $\cos \omega_0 t = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$

$$x(t) \cdot \cos \omega_0 t = \frac{1}{2} [x(t) \cdot e^{j\omega_0 t} + x(t) \cdot e^{-j\omega_0 t}]$$

$$FT \{ x(t) \cdot \cos \omega_0 t \} = \frac{1}{2} [FT \{ x(t) \cdot e^{j\omega_0 t} \} + FT \{ x(t) \cdot e^{-j\omega_0 t} \}]$$

$$= \frac{1}{2} [X(j(\omega - \omega_0)) + X(j(\omega + \omega_0))]$$

ii) $\sin \omega_0 t = \frac{j}{2} [e^{j\omega_0 t} - e^{-j\omega_0 t}]$

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Fourier Transform Properties PART 6

10. Multiplication in Time:

$$x_1(t) \cdot x_2(t) \Leftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(j\omega) X_2(j\omega) e^{j\omega t} d\omega$$

11. Differentiation in Time:

$$\frac{d}{dt} x(t) \Leftrightarrow (j\omega) X(j\omega)$$

12. Integration in Time:

$$\int_{-\infty}^t x(\tau) d\tau \Leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(j0) \delta(\omega)$$

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Properties of Fourier Transform (Part 6)

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Fourier Transform Properties PART 7

11. Differentiation in Time:

$$\frac{d}{dt} x(t) \Leftrightarrow (j\omega) X(j\omega)$$

12. Integration in Time:

$$\int_{-\infty}^t x(\tau) d\tau \Leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(j0) \delta(\omega)$$

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Properties of Fourier Transform (Part 7)

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Fourier Transform Properties PART 8

13. Differentiation in Frequency:

$$t \cdot x(t) \Leftrightarrow (j) \frac{d}{d\omega} X(j\omega)$$

14. Modulation:

$$x(t) \cdot \cos \omega_0 t \Leftrightarrow \frac{1}{2} [X(j(\omega + \omega_0)) + X(j(\omega - \omega_0))]$$

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Properties of Fourier Transfo...

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Parseval's Energy Theorem

15. Parseval's Energy Theorem:

$$E_{x(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

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Parseval's Energy Theorem

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Fourier Transform (Duality Property)

16. Duality Property:

$$X(j\omega) \Leftrightarrow x(t)$$

$$x(t) \Leftrightarrow X(j\omega)$$

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Duality Property of Fourier Transform

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Fourier Transform

Fourier Transform of Basic

Fourier Transform of Basic Signals (Impulse Signal)

$$x(t) = \delta(t) \Rightarrow X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t-0) \cdot e^{-j\omega t} dt$$

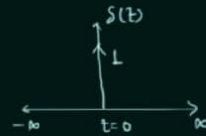
$$= \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega \cdot 0} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) dt$$

$$X(\omega) = 1$$

$$\delta(t) \Rightarrow 1$$

$$x(t_1) \cdot \delta(t-t_0) = x(t_0) \cdot \delta(t-t_0)$$



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Fourier Transform (Duality Property)

16. Duality Property:

$$x(t) \Rightarrow X(\omega)$$

$$X(t) \Rightarrow 2\pi x(-\omega)$$

$$f(t) \Rightarrow F(\omega) \Rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

6:17

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Duality Property of Fourier Transform

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Fourier Transform (DC Value)

$$x(t) \Rightarrow X(\omega)$$

$$X(0) = \int_{-\infty}^{\infty} x(t) dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi x(t) dt$$

$$= \frac{A}{2\pi} \int_{-\infty}^{\infty} \delta(t) dt$$

$$= \frac{A}{2\pi} \cdot 2\pi = A$$

$$X(0) = \frac{A}{2\pi} \int_{-\infty}^{\infty} \delta(t) dt$$

8:52

186/Signal & System

Fourier Transform of Basic Signals (DC Value)

Neso Academy

1.8 lakh views · 6 years ago

Fourier Transform (Impulse Signal)

$$x(t) = \delta(t) \Rightarrow X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega \cdot 0} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) dt$$

$$= 1$$

Impulse related signals are absolutely integrable signals and hence we can use the formula of Fourier Transform

...

187/Signal & System

Fourier Transform of Basic S...

Neso Academy

1.7 lakh views · 6 years ago



Fourier Transform (Exponential Signal)

$$x(t) = e^{at} u(t)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{at} u(t) e^{-j\omega t} dt$$

$$= \int_{0}^{\infty} e^{at} e^{-j\omega t} dt$$

$$= \int_{0}^{\infty} e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{0}^{\infty}$$

$$= \frac{e^{(a-j\omega)t}}{a-j\omega} \bigg|_{0}^{\infty}$$

$$= \frac{e^{(a-j\omega)t}}{a-j\omega} \bigg|_{0}^{\infty}$$

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$$= \frac{e^{(a-j\omega)t}}{a-j\omega} \bigg|_{0}^{\infty}$$

$$= \frac{e^{(a-j\omega)t}}{a-j\omega} \bigg|_{0}^{\infty}$$

$$= \frac{e^{(a-j\omega)t}}{a-j\omega} \bigg|_{0}^{\infty}$$

9:06

188/Signal & System

Fourier Transform of Basic Signals (Exponential Signals)

Neso Academy

2.3 lakh views · 6 years ago

Fourier Transform (Signum Function)

$$x(t) = \text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt$$

8:03

189/Signal & System

Fourier Transform of Basic Signals (Signum Function)

Neso Academy

1.7 lakh views · 6 years ago

$$\underline{x(t)} = \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$$

$$T_0 = 4 \text{ sec}$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cdot \cos n\omega_0 t dt$$

$$\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{4}$$

$$a_n = \frac{1}{2} \int_{-1}^3 x(t) \cdot \cos \frac{n\pi}{2} t dt$$

$$\omega_0 = \frac{\pi}{2} \text{ rad/sec}$$

$$a_n = \frac{1}{2} \left[\int_{-1}^1 (1) \cos \frac{n\pi}{2} t dt + \int_1^3 (-1) \cdot \cos \frac{n\pi}{2} t dt \right]$$

$$a_n = \frac{1}{2} \left[\int_{-1}^1 \cos \frac{n\pi}{2} t dt - \int_1^3 \cos \frac{n\pi}{2} t dt \right]$$

$$a_n = \frac{1}{2} \left[\int_{-\pi/2}^{\pi/2} \cos \theta \right]$$

$$\frac{n\pi}{2} t = \theta$$

$$\frac{n\pi}{2} dt = d\theta \Rightarrow dt = \frac{2}{n\pi} d\theta$$

$$t = -1 \Rightarrow \theta = -\frac{n\pi}{2}$$

$$t = 1 \Rightarrow \theta = \frac{n\pi}{2}$$

$$t = 3 \Rightarrow \theta = \frac{3n\pi}{2}$$

$$a_n = \frac{1}{2} \times \frac{1}{n\pi} \left[\int_{-n\pi/2}^{n\pi/2} \cos \theta d\theta - \int_{n\pi/2}^{3n\pi/2} \cos \theta d\theta \right]$$

$$a_n = \frac{1}{n\pi} \left[(\sin \theta)_{-n\pi/2}^{n\pi/2} - (\sin \theta)_{n\pi/2}^{3n\pi/2} \right]$$

$$a_n = \frac{1}{n\pi} \left[\sin \frac{n\pi}{2} - \sin(-\frac{n\pi}{2}) - \sin \frac{3n\pi}{2} + \sin \frac{n\pi}{2} \right]$$

Case I :- $n = \text{even}$

$$a_n = \frac{1}{n\pi} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} - \sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right]$$

$$a_n = \frac{2}{n\pi} \sin \frac{n\pi}{2}$$

Case II :- $n = \text{odd}$

$$a_n = \frac{1}{n\pi} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} - \sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right]$$

$$a_n = \frac{2}{n\pi} \sin \frac{n\pi}{2}$$

Fourier Series | Chapter-4... · 4/33

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142 Signal & System 13:38

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Trigonometric Fourier Series
When $x(t)$ is purely even there will be no sine term as sine is an odd signal and this means $b_n = 0$

143 Signal & System 13:34

Trigonometric Fourier Series
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Trigonometric Fourier Series (Example 1)

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Trigonometric Fourier Series...
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Trigonometric Fourier Series (Example 2)

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Trigonometric Fourier Series (Example 2)
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Trigonometric Fourier Series (Example 3)
ESE 2011

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Trigonometric Fourier Series (Example 3)
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Trigonometric Fourier Series (Example 4)
GATE 2006

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Trigonometric Fourier Series (Example 4)
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Complex Exponential Fourier Series

$$\checkmark x(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\omega_0 t}$$

For any real time domain signal, the exponential Fourier series coefficient (C_n) will be conjugate symmetric and vice versa is also true

$C_n \rightarrow$ Complex Exp. Fourier Coeff

if C_n is CS :-

$$\checkmark C_n = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{-jn\omega_0 t} dt \quad \text{--- (I)}$$

$$C_n = C_{-n}^* \Rightarrow x(t) = x^*(t) \Rightarrow \underline{\underline{\text{Real}}}$$

↓ Reversal ($n = -n$)

$$C_{-n} = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{jn\omega_0 t} dt$$

↓ *

$$C_{-n}^* = \frac{1}{T_0} \int_{T_0} \underbrace{x^*(t)} \cdot e^{-jn\omega_0 t} dt \quad \text{--- (II)}$$

C_n 

$$C_n = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{-jn\omega_0 t} dt \quad \text{--- (i)}$$

↓ Reversal ($n = -n$)

$$C_{-n} = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{jn\omega_0 t} dt$$

↓ *

$$C_{-n}^* = \frac{1}{T_0} \int_{T_0} x^*(t) \cdot e^{-jn\omega_0 t} dt \quad \text{--- (ii)}$$

→ $x(t)$ is real $\Rightarrow C_n$ is CS

$|C_n| \Rightarrow \text{Even}$
 $\angle C_n \Rightarrow \text{Odd}$

$$C_n = |C_n| e^{j\angle C_n} \quad \text{--- (iii)}$$

↓ $n = -n$

$$C_{-n} = |C_{-n}| e^{j\angle C_{-n}}$$

↓ *

$$C_{-n}^* = |C_{-n}| e^{-j\angle C_{-n}} \quad \text{--- (iv)}$$

$$|C_n| e^{j\angle C_n} = |C_{-n}| e^{-j\angle C_{-n}}$$

$$|C_n| = |C_{-n}| \Rightarrow \text{Even}$$

$$\angle C_n = -\angle C_{-n}$$

$$-\angle C_n = \angle C_{-n} \Rightarrow \text{Odd}$$

Ex :

Fourier Series | Chapter-4... · 9/33

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Trigonometric Fourier Series (Example 4)

Question: $x(t)$ is a real valued function of a real variable with period T . Its trigonometric Fourier series expansion contains no terms of frequency $\omega = 2\pi(20/7), \omega = 1, 2, \dots$. Also, no odd terms are present. Then $x(t)$ satisfies the equation.

- (a) $x(t) = x(t-T)$
- (b) $x(t) = x(T-t) = x(-t)$
- (c) $x(t) = x(T-t) = x(t-T/2)$
- (d) $x(t) = x(t-T) = x(t-T/2)$

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8:59

147 Signal & System

Trigonometric Fourier Series (Example 4)

Neso Academy

1.3 lakh views · 6 years ago

Trigonometric Fourier Series (Example 5)

Question: Prove $x(t)$ is periodic

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Condition for periodicity: $x(t) = x(t \pm T)$

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

7:54

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Trigonometric Fourier Series (Example 5)

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1.1 lakh views · 6 years ago

Exponential Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$C_n \rightarrow \text{Complex Exp. Fourier Coeff.}$

$$C_n = \frac{1}{T} \int_{T_0} x(t) e^{-jn\omega_0 t} dt$$

For any real time domain signal, the exponential Fourier series coefficient (C_n) will be conjugate symmetric and vice versa is also true

149 Signal & System

Complex Exponential Fourier...

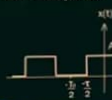
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Exponential Fourier Series (Example 1)

Question: Find C_n for signal $x(t)$



13:03

150 Signal & System

Complex Exponential Fourier Series (Example 1)

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Exponential Fourier Series (Example 2)

Question: The Fourier series representation of an impulse train denoted by

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

GATE 1999

$$(a) \frac{1}{T_0} \sum_{n=-\infty}^{\infty} \exp\left\{-\frac{j2\pi n t}{T_0}\right\}$$

$$(c) \frac{1}{T_0} \sum_{n=-\infty}^{\infty} \exp\left\{\frac{j2\pi n t}{T_0}\right\}$$

$$(b) \frac{1}{T_0} \sum_{n=-\infty}^{\infty} \exp\left\{-\frac{j2\pi n t}{T_0}\right\}$$

$$(d) \frac{1}{T_0} \sum_{n=-\infty}^{\infty} \exp\left\{\frac{j2\pi n t}{T_0}\right\}$$

8:58

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Complex Exponential Fourier Series (Example 2)

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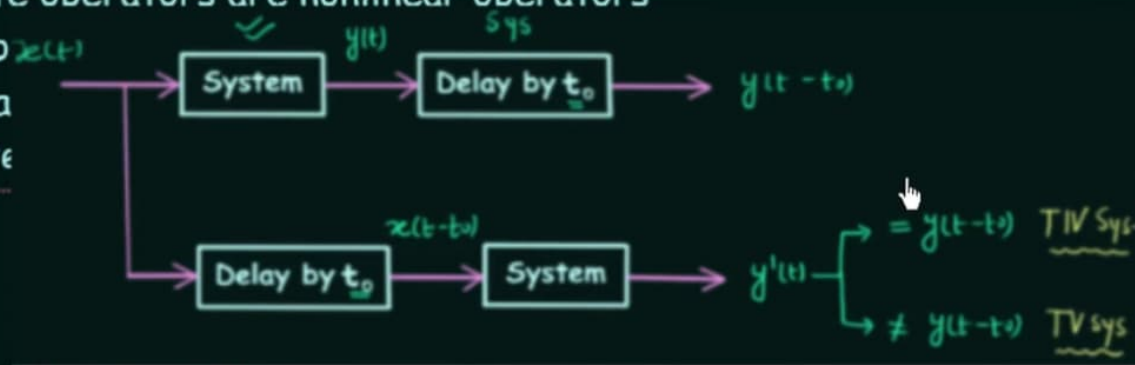
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Prerequisites for LTI Systems (Revision of Linearity & Time Variance)

- >> System linearity is independent of time scaling
- >> System linearity is independent of coefficient used in system relationship
- >> If any added/subtracted term other than i/p and o/p is available in the system relationship then the system will be nonlinear
- >> If output is summation of time shifted terms of input, then the system will be linear
- >> Integral and Differential operators are linear operators
- >> Even and Odd operators are linear operators
- >> Real, Imaginary and Conjugate operators are nonlinear operators
- >> Trigonometric, Inverse Trigonometric, sa, sinc,
- >> For zero i/p, o/p is also equal to zero
- >> Split systems are linear systems

For TIV System:

- >> No time scaling
- >> Coefficient in system relationship should be constant
- >> Any added/subtracted term in the system relationship (except i/p and o/p) must be constant or zero



and if the output of the system is same as the output recorded here we will call

Prerequisites for LTI Systems (Revision of Linearity & Time Variance)

- » System linearity is independent of time scaling
- » System linearity is independent of coefficient used in system relationship
- » If any added/subtracted term other than i/p and o/p is available in the system relationship then the system will be nonlinear
- » If output is summation of time shifted terms of input, then the system will be linear
- » Integral and Differential operators are linear operators
- » Even and Odd operators are linear operators
- » Real, Imaginary and Conjugate operators are nonlinear operators
- » Trigonometric, Inverse Trigonometric, Logarithmic, Exponential, Roots, Powers, Modulus, sign, sinc,
- » For zero i/p, o/p is also equal to zero
- » Split systems are linear systems

For TIV System:

- » No time scaling
- » Coefficient in system relationship should be constant
- » Any added/subtracted term in the system relationship (except i/p and o/p) must be constant or zero

$$1.4y(t) = 1.2x(t) + 5$$

↑ constant
TIV + LTI → X LTI

Prerequisites for LTI Systems (Revision of Linearity & Time Variance)

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1.3K

36



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Remix



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Prerequisites for LTI Systems (Laplace Transform)

Time Domain:

$$y(t) = x(t-1) + x(t+1)$$

$$\begin{array}{c} \uparrow \text{ILT} \\ \downarrow \text{LT} \end{array}$$

Freq. Domain:

$$Y(s) = X(s)e^{-s} + X(s)e^{s}$$

$$L\{f(t)\} = F(s) = \int_{-\infty}^{\infty} f(t) \cdot e^{-st} dt$$

Complex Complex

IIO / Signal & System

14:48



Next: Prerequisites for LTI Systems (Laplace Tra...

LTI system neso academy · 1/8

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imp

Properties of Convolution (Part 1)

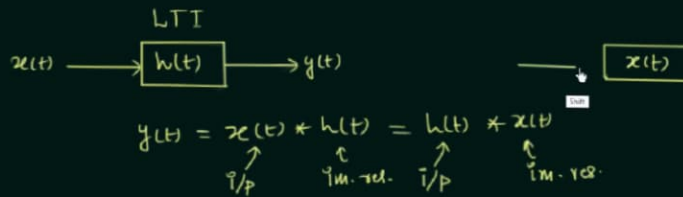
(10 properties)

1. Commutative Property:

$$x_1(t) * x_2(t) = y(t) = x_2(t) * x_1(t)$$

$$\downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow$$

$$\underbrace{x_1(\tau)}_{\text{fix}} \quad \underbrace{x_2(t-\tau)}_{\text{move}} \quad \underbrace{x_2(\tau)}_{\text{fix}} \quad \underbrace{x_1(t-\tau)}_{\text{move}}$$



Chapters



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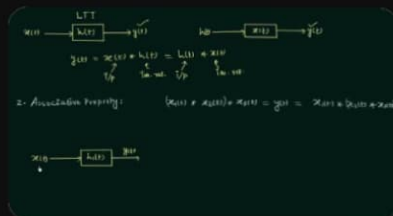


Properties of Convolution

Commutative Property

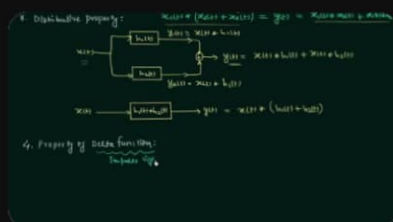


0:00



Associative Property

6:20



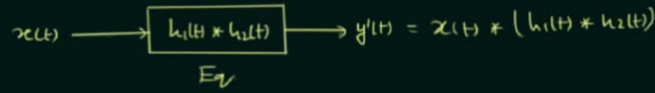
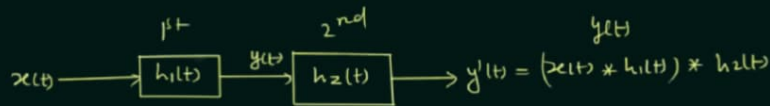
Property of Delta Function

13:32

~ i/p i/p i/p i/p

2. Associative Property:

$$(x_1(t) * x_2(t)) * x_3(t) = y(t) = x_1(t) * (x_2(t) * x_3(t))$$



3. Distributive Property:

$$x_1(t) * (x_2(t) + x_3(t)) = y(t) = x_1(t) * x_2(t) + x_1(t) * x_3(t)$$

$h_1(t)$

Chapters



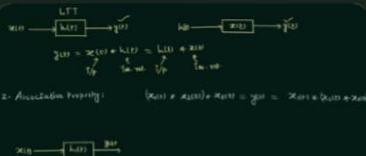
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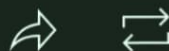
Properties of Convolution

Commutative Property

0:00



Associative Property

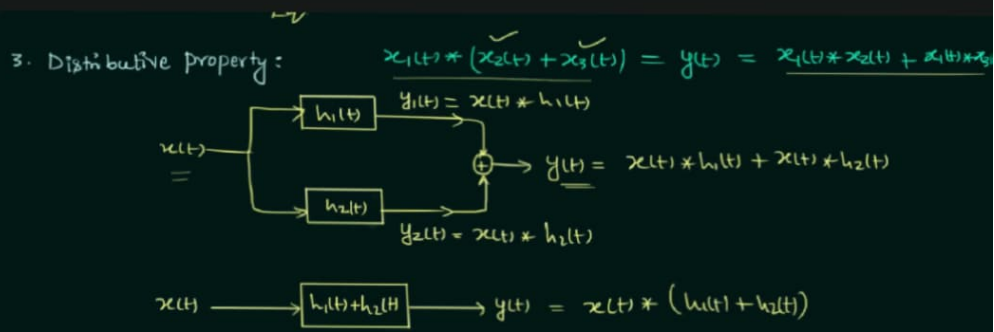


6:20



Property of Delta Function

13:32



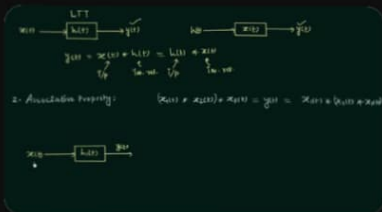
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Properties of Convolution

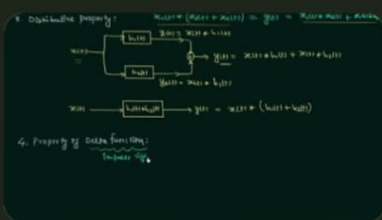
Commutative Property

0:00



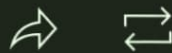
Associative Property

6:20



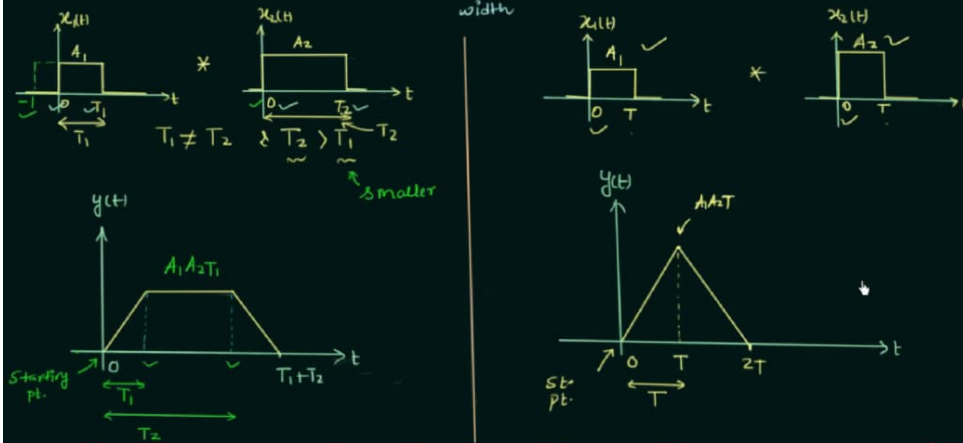
Property of Delta Function

13:32



Convolution Operation (Shortcut)

- >> Convolution of two rectangular pulses of unequal duration will be a trapezoid ✓
- >> Convolution of two rectangular pulses of equal duration will be a triangle ✓



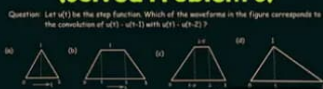
Signals and Systems • 120/313

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124/Signal & System

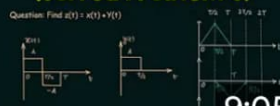
Convolution (Solved Problem 5)



14:49

125/Signal & System

Convolution (Solved Problem 6)



9:05

126/Signal & System

Static LTI System

Condition

$$y(t) = kx(t)$$

$$Y(s) = kX(s) \Rightarrow \frac{Y(s)}{X(s)} = k$$

$$H(s) = k$$

$$1. \frac{y(t)}{x(t)} = \frac{x(t+1)}{x(t)} \rightarrow \frac{Y(s)}{X(s)} = e^{s} \rightarrow H(s) = e^{s}$$

11:24

127/Signal & System

Static LTI System (Solved Problems)

$$1. \frac{y(t)}{x(t)} = \frac{x(t-1)}{x(t)} \rightarrow \frac{Y(s)}{X(s)} = e^{-s} \rightarrow H(s) = e^{-s}$$

$$2. \frac{y(t)}{x(t)} = \frac{x(t+1)}{x(t)} \rightarrow \frac{Y(s)}{X(s)} = e^{s} \rightarrow H(s) = e^{s}$$

8:46

128/Signal & System

Causal LTI System

$$y(t) = \int_{-\infty}^t x(\tau) d\tau \Rightarrow \frac{Y(s)}{X(s)} = \frac{1}{s} \rightarrow H(s) = \frac{1}{s}$$

8:00

129/Signal & System

Convolution (Solved Problem 5)

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Convolution (Solved Problem 6)

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Static LTI System

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Static LTI System (Solved Problems)

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Causal LTI System

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1 lakh views • 6 years ago

$$E = P \times T$$

↑ ↑
finite × ∞

properties: 1) P.S. are power signals but vice-versa is not true

$$2) P = (RMS)^2 \Rightarrow RMS = \sqrt{P}$$



Power Signals

1.9 lakh views 7 yr ago #SignalAndSystemByNeso ...more



Neso Academy 26.4 lakh



1.7K

55



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Remix



Do

Comments 113

M

Though it is a periodic signal, why are we taking the limits as $-t/2$ to $t/2$ like non periodic signals? we need...

Power Signals (Solved Problems)

PART
1

Question: Calculate average power of following signals

$$1) x_1(t) = A_0 \sin \omega_0 t$$

$$P = \int_{-T_0/2}^{T_0/2} \dots$$

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A_0^2 \sin^2 \omega_0 t \, dt$$

$$P = \frac{A_0^2}{T_0} \int_{-T_0/2}^{T_0/2} \left(\frac{1 - \cos 2\omega_0 t}{2} \right) dt$$

$$2) x_2(t) \rightarrow P = 4$$

$$y(t) = 4x_2(2t + 4)$$

$$y(t) \rightarrow P = ?$$

45

Signal & System

12:25



Power of CT Signals (Solved Problems) I



Next: Power of CT Signals (Solved Problems) | P...

Signals and Systems • 44/313 7 years ago

Time Scaling $\rightarrow x_2(t) = x_1(2t)$
 $= A_0 \sin 2\omega_0 t$ $\left\{ \begin{array}{l} P = A_0^2/2 \\ RMS = A_0/\sqrt{2} \end{array} \right.$

Phase Shift $\rightarrow x_3(t) = A_0 \sin(\omega_0 t + \phi)$ $\left\{ \begin{array}{l} P = A_0^2/2 \text{ \& } RMS = A_0/\sqrt{2} \end{array} \right.$

Time Rev. $\rightarrow x_4(t) = x_1(-t)$
 $= A_0 \sin(-\omega_0 t) = \underline{\underline{-A_0 \sin \omega_0 t}}$ $\left\{ \begin{array}{l} P \text{ \& } RMS \text{ will remain same} \end{array} \right.$

Time Shifting $\rightarrow x_5(t) = x_1(t+2)$
 $= A_0 \sin[\omega_0(t+2)]$
 $= A_0 \sin(\omega_0 t + 2\omega_0)$ $\left\{ \begin{array}{l} P = A_0^2/2 \\ RMS = A_0/\sqrt{2} \end{array} \right.$

Amp. Rev. $\rightarrow x_6(t) = -x_1(t)$
 $= \underline{\underline{-A_0 \sin \omega_0 t}}$ $\left\{ \begin{array}{l} P = A_0^2/2 \\ RMS = A_0/\sqrt{2} \end{array} \right.$

Amp. Scaling $\rightarrow x_7(t) = 2$

$$1) x_1(t) = A_0 \sin \omega_0 t$$

↓
periodic $\Rightarrow$ Power signal

$$P = \text{finite}$$

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A_0^2 \sin^2 \omega_0 t \, dt$$

$$P = \frac{A_0^2}{T_0} \int_{-T_0/2}^{T_0/2} \left(\frac{1 - \cos 2\omega_0 t}{2} \right) dt$$

$$P = \frac{A_0^2}{2T_0} \left[\int_{-T_0/2}^{T_0/2} 1 \, dt - \int_{-T_0/2}^{T_0/2} \cos 2\omega_0 t \, dt \right] = \frac{A_0^2}{2}$$

$$RMS = \sqrt{P} = \sqrt{\frac{A_0^2}{2}} = A_0 / \sqrt{2}$$

$$2) x_g(t) \rightarrow P = 4$$

$$y(t) = 4j x_g(2t+4)$$

$$y(t) \rightarrow P = ?$$

$$x_g(t) \rightarrow 4$$

$$x_g(t+4) \rightarrow 4$$

$$x_g(2t+4) \rightarrow 4$$

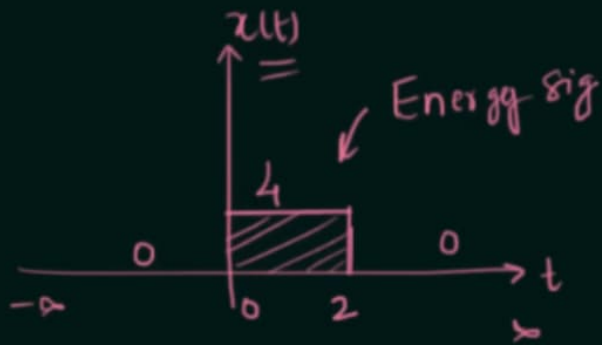
$$y(t) = \underbrace{4j}_k x_g(2t+4) \rightarrow |k|^2 \times 4$$

$$4^2 \times 4$$



Energy Signals

$$E = \text{finite} \Rightarrow P$$



$$\begin{aligned}
 \underline{E} &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\
 &= \int_{-\infty}^0 0 dt + \int_0^2 16 dt + \int_2^{\infty} 0 dt \\
 E &= [16t]_0^2 = 32 \text{ J (finite)}
 \end{aligned}$$

av. value of finite duration signal = 0

$$\begin{aligned}
 \checkmark \text{ Av. power} &= \frac{\text{total power}}{\text{total time}} \rightarrow \\
 &= \text{av. to av} \\
 &= T = \infty
 \end{aligned}$$

$$P = \lim_{T \rightarrow \infty} \frac{E}{T}$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} |x(t)|^2 dt &= \text{finite} \\
 \downarrow \\
 \int_{-\infty}^{\infty} |x(t)|^2 dt &= \text{finite}
 \end{aligned}$$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^0 0 dt + \int_0^2 16 dt + \int_2^{\infty} 0 dt$$

$$\underline{E} = [16t]_0^2 = \underline{32 \text{ J (finite)}}$$

$-\infty$ to ∞

$$\underline{T = \infty}$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$\underline{P = \lim_{T \rightarrow \infty} \frac{E}{T}}$$

1. E.S. are absolutely integrable signal

F.T.

$$\int_{-\infty}^{\infty} |x(t)| dt = \text{finite} < \infty$$

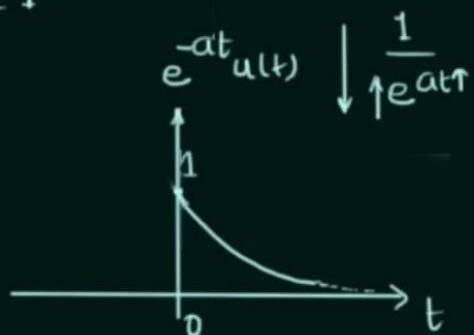
2. E = area under $|x(t)|^2$ graph

$$3. P = \lim_{T \rightarrow \infty} \frac{E}{T}$$

$$e^{-at}, a > 0$$

→ $\oplus^{\infty}, t < 0$

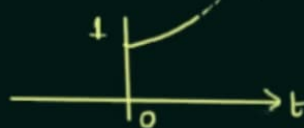
$$e^0 = 1$$



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} (e^{-at}u(t))^2 dt$$

$$c-\Pi \quad e^{-at}u(t), a < 0 [-\infty]$$

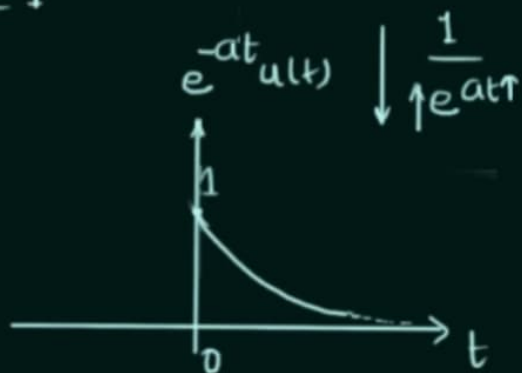


$$\boxed{E = A_0} = \text{Area under } |x(t)|^2$$

$E = \int_{-\infty}^{\infty} x^2$

$\hookrightarrow \Theta^{\text{re}}, t < 0$

$$e^0 = 1$$



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

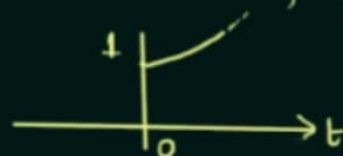
$$= \int_{-\infty}^{\infty} (e^{-at}u(t))^2 dt$$

$$= \int_{-\infty}^0 0 dt + \int_0^{\infty} (e^{-at} \cdot 1)^2 dt$$

$$= \int_0^{\infty} -2at \cdot e^{-2at} dt$$



$$C-II \quad e^{-at}u(t), a < 0 [-\infty]$$

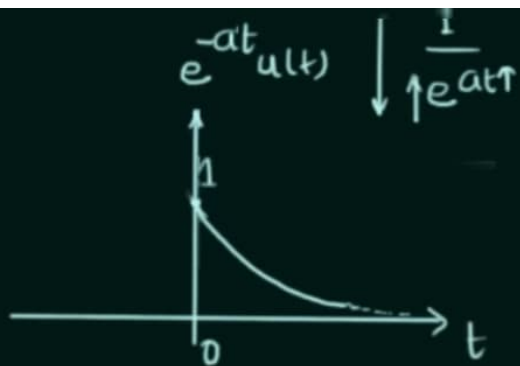


$$\boxed{E = \infty} = \text{Area under } |x(t)|^2$$

$$E \cdot s \cdot x$$

$$\int_x^y e^{at} = \frac{1}{a} [e^{at}]_x^y$$

$$C-III \quad e^{0t}u(t), a = 0$$



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

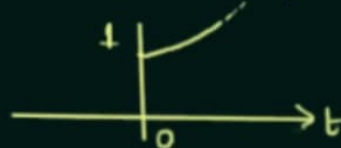
$$= \int_{-\infty}^{\infty} (e^{-at}u(t))^2 dt$$

$$= \int_{-\infty}^0 0 dt + \int_0^{\infty} (e^{-at})^2 dt$$

$$= \int_0^{\infty} e^{-2at} dt \quad \leftarrow \oplus \infty$$



$$C-II \quad e^{-at}u(t), \quad a < 0 \quad [-\infty]$$

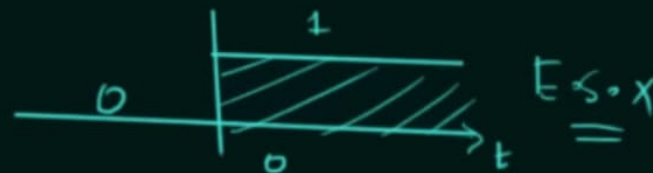


$$\boxed{E = \infty} = \text{Area under } |x(t)|^2$$

$E.S.X$

$$\int_x^y e^{at} = \frac{1}{a} [e^{at}]_x^y$$

$$C-III \quad e^{0t}u(t), \quad a=0$$



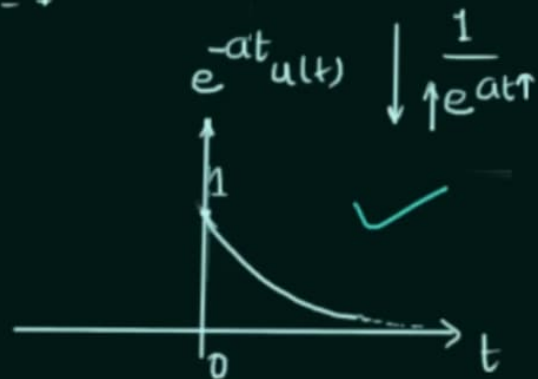
Question: Calculate total energy of the following signals:

1) $x_1(t) = e^{-at} \boxed{u(t)}$, $a > 0$ (+ve)

$\mathbb{C}^{-1}$

e^{-at} $\rightarrow \ominus^ve$, $t > 0$
 $\rightarrow \oplus^ve$, $t < 0$

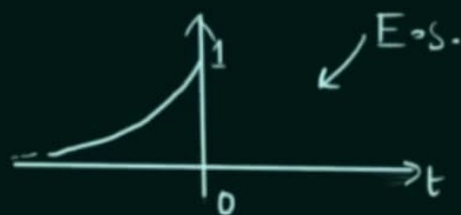
$e^0 = 1$



$$E = \int_{-\infty}^{\infty} |x_1(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} (e^{-at}u(t))^2 dt$$

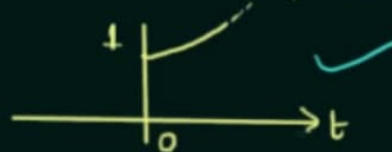
2) $x_2(t) = x_1(-t)$



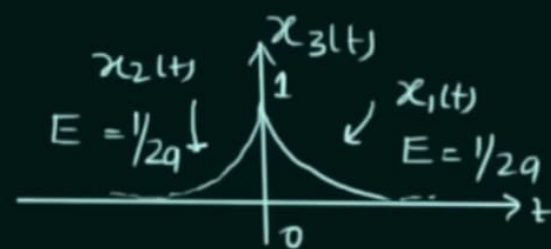
$E = 1/2a$

No effect of time rev.
of total energy

$\mathbb{C}^{-II}$ $e^{-at}u(t)$, $a < 0$ [-ve]



3) $x_3(t) = x_1(t) + x_2(t)$



$E = 1/2a + 1/2a$

$= 1/a$

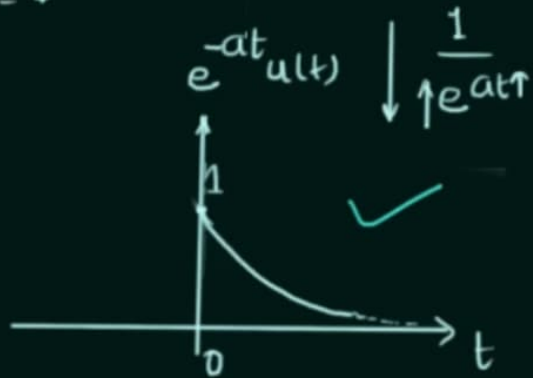
Ans

Question: Calculate total energy of the following signals:

1) $x_1(t) = e^{-at} \boxed{u(t)}$, $a > 0$ (+ve)

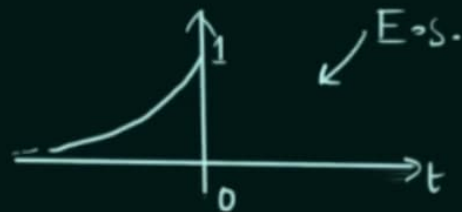
$\checkmark$ e^{-at} $\rightarrow \Theta^+$, $t > 0$
 $\checkmark$ e^{-at} $\rightarrow \Theta^+$, $t < 0$

$e^0 = 1$



$\checkmark E = \int_{-\infty}^{\infty} |x_1(t)|^2 dt$
 $= \int_{-\infty}^{\infty} (e^{-at} u(t))^2 dt$

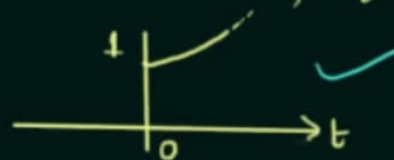
2) $x_2(t) = x_1(-t)$



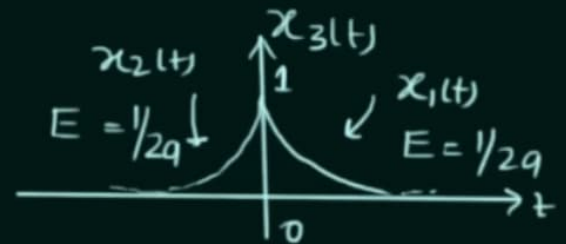
$E = 1/2a$

No effect of time rev.
of total energy

$\checkmark$ $e^{-at} u(t)$, $a < 0$ [-ve]



3) $x_3(t) = x_1(t) + x_2(t)$



$E = 1/2a + 1/2a$
 $= 1/a$ Ans

$x_1(t) \rightarrow E_1$

$x_2(t) \rightarrow E_2$

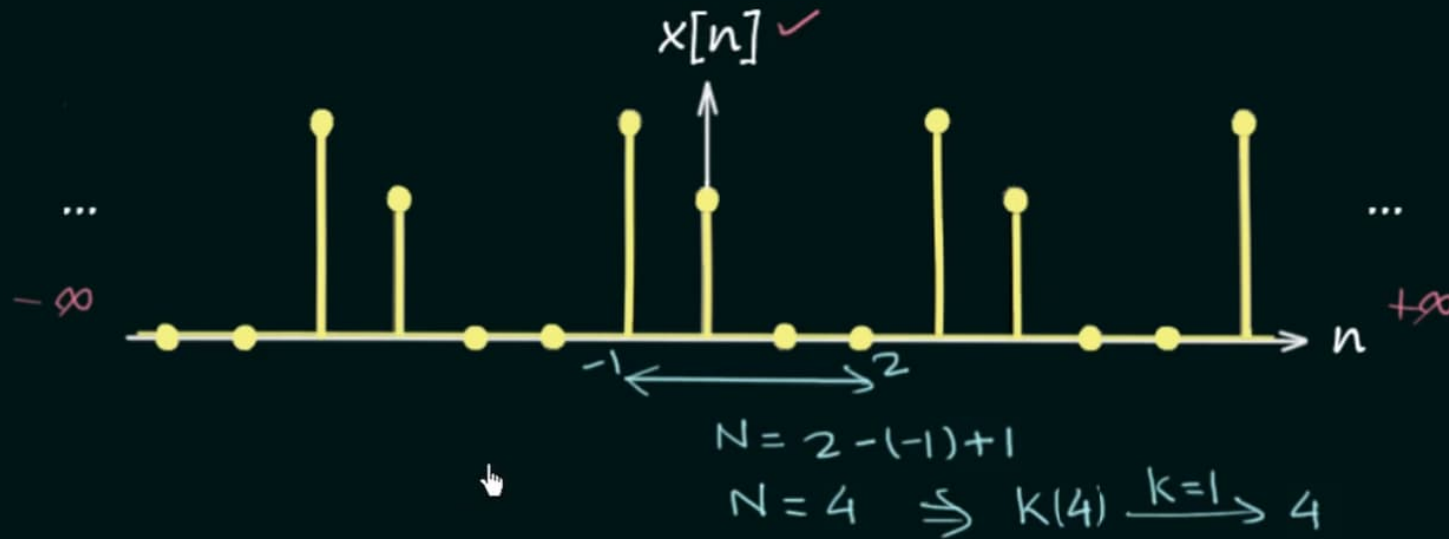
$x_3(t) = \underline{x_1(t)} + \underline{x_2(t)}$
 $E_3 = E_1 + E_2$

✓ Periodic Discrete Time Signals

kN
 $\swarrow \searrow$
 $\text{int.} \quad \text{FTP(int.)}$

$$x[n] = x[n \pm kN]$$

$\downarrow \uparrow$
 $\boxed{N}$ P



the shifting you will have the same signal and this is happening

Discrete-Time Energy Signals

$$\rightarrow x(t)$$
$$E = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

$$\rightarrow x[n]$$
$$E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$$

For both Periodic & Non-Periodic
Signals

✓ Discrete-Time Energy Signals

→ $x(t)$

$$E = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

→ $E = \text{finite} \Rightarrow P = 0$

→ $x[n]$

$$E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$$

→ Energy signals $\Rightarrow$ Ab. int. $\rightarrow \int_{-\infty}^{\infty} |x(t)| dt < \infty$

$\Longleftrightarrow$ Ab. summable

Example: $x[n] = \{1, 2, 3, 4, 5\}$

Find total energy of signal $x[n]$.

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

Discrete-Time Energy Signals (Solved Problems)

1. $x[n] = (1/3)^n u[n]$ $E = ?$ $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2 = \sum_{n=0}^{+\infty} (1/3)^{2n} = \sum_{n=0}^{+\infty} (1/9)^n = \frac{1}{1 - 1/9} = \frac{9}{8} \text{ J}$

2. $x[n] = \delta[n]$ $E = ?$ $E = \sum_{n=-\infty}^{+\infty} |\delta[n]|^2 = 1 \text{ J}$

3. $x[n] = \{1+j, 1-j, -2, 2\}$ $E = ?$ $E = 1^2 + 1^2 + (-2)^2 + 2^2 = 10 \text{ J}$

NOTE: There is no effect of Time Shifting, Time Reversal, Time Expansion & Amplitude Reversal on the total energy of the DTS.

$\rightarrow x[n] = \{1, 2, 3, 4, 5\} \rightarrow E = 55 \text{ J}$ $\text{ii) } x[-n] \rightarrow E = 55 \text{ J}$ $\text{iv) } x[3n] \rightarrow E = 17 \text{ J}$

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Discrete Energy Signals

$E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$ $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$ $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$

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Discrete Time Energy Signals

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Discrete Energy Signals (Solved Problems)

$E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$ $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$ $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$

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Discrete Time Energy Signal...

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Discrete Power Signals

$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$ $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$ $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$

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Discrete Time Power Signals

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Discrete Power Signals (Solved Problems)

$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$ $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$ $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$

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Discrete Time Power Signals (Solved Problems)

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Unit Impulse Sequence

$\delta[n] = 1, n=0; 0, \text{ elsewhere}$ $\delta[n] = 1, n=0; 0, \text{ elsewhere}$ $\delta[n] = 1, n=0; 0, \text{ elsewhere}$

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Unit Impulse Sequence

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Unit Step & Ramp Sequences

Unit Step & Unit Ramp

Discrete-Time Power Signals

Average Power of CTS $x(t)$

$$P = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt, \quad x(t) \text{ is periodic}$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} |x(t)|^2 dt, \quad x(t) \text{ is NP}$$

$T/2 - (-T/2) = T$

Average Power of DTS $x[n]$

$$P = \frac{1}{N} \sum_{n=-N}^N |x[n]|^2, \quad x[n] \text{ is periodic}$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |x[n]|^2, \quad x[n] \text{ is NP}$$

$N - (-N) = 2N + 1$

****** $\rightarrow P = \text{finite} \Rightarrow E = \infty$

$0 < P < \infty$

Example: $x[n] = \cos \frac{\pi}{2} n$, Calculate average power of signal $x[n]$.

ω_0

Sol.:- $x(t) = \cos \frac{\pi}{2} t \rightarrow \text{periodic}$

$\rightarrow \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/2} = 4 \text{ (rat.)} \Rightarrow x[n] \text{ is Per.}$

$\rightarrow N = \frac{2\pi}{\omega_0} K \Rightarrow N = 4K \Rightarrow \underline{N=4}$

$$\left\{ \begin{array}{l} P = \frac{1}{4} \sum_{n=0}^3 |\cos \frac{\pi}{2} n|^2 \\ P = \frac{1}{4} \{ |\cos 0|^2 + |\cos \frac{\pi}{2}|^2 + |\cos \pi|^2 + |\cos \frac{3\pi}{2}|^2 \} \end{array} \right.$$

Discrete-Time Power Signals

Average Power of CTS $x(t)$

$$P = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt, \quad x(t) \text{ is periodic}$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} |x(t)|^2 dt, \quad x(t) \text{ is NP}$$

$T/2 - (-T/2) = T$

Average Power of DTS $x[n]$

$$P = \frac{1}{N} \sum_{n=-N}^N |x[n]|^2, \quad x[n] \text{ is periodic}$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |x[n]|^2, \quad x[n] \text{ is NP}$$

$N - (-N) = 2N + 1$

****** $\rightarrow P = \text{finite} \Rightarrow E = \infty$

$$0 < P < \infty$$

Example: $x[n] = \cos \left[\frac{\pi}{2} n \right]$, Calculate average power of signal $x[n]$.

Sol.:- $x(t) = \cos \frac{\pi}{2} t \rightarrow \text{periodic}$

$\rightarrow \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/2} = 4 \text{ (rat.)} \Rightarrow x[n] \text{ is Per.}$

$\rightarrow N = \frac{2\pi}{\omega_0} K \Rightarrow N = 4K \Rightarrow \underline{N=4}$

$$\left\{ \begin{array}{l} P = \frac{1}{4} \sum_{n=0}^3 \left| \cos \frac{\pi}{2} n \right|^2 \\ P = \frac{1}{4} \left\{ \left| \cos 0 \right|^2 + \left| \cos \frac{\pi}{2} \right|^2 + \left| \cos \pi \right|^2 + \left| \cos \frac{3\pi}{2} \right|^2 \right\} \\ P = 1/2 \quad \text{Ans} \end{array} \right.$$

Sol :-

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |x[n]|^2$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |A_0 u[n]|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=0}^N |A_0|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=0}^N A_0^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \underbrace{\{A_0^2 + A_0^2 + \dots + A_0^2\}}_{\substack{n=1 \text{ to } \\ n=N \\ NA_0^2}}$$

$$= \lim_{N \rightarrow \infty} \frac{A_0^2(1+N)}{(2N+1)}$$

$$= \lim_{N \rightarrow \infty} \frac{A_0^2(1/N+1)}{N(2+1/N)}$$

$$= \frac{A_0^2(0+1)}{1(2+0)} \Rightarrow P = \frac{A_0^2}{2}$$



Sol. :-

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |x[n]|^2$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |A_0 u[n]|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=0}^N |A_0|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=0}^N A_0^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \left\{ A_0^2 + A_0^2 + \dots + A_0^2 \right\}$$

$\underbrace{\hspace{10em}}_{\substack{n=1 \text{ to} \\ n=N \\ NA_0^2}}$

$$= \lim_{N \rightarrow \infty} \frac{A_0^2 (1+N)}{(2N+1)}$$

$$= \lim_{N \rightarrow \infty} \frac{A_0^2 (1/N + 1)}{2 + 1/N}$$

$$= \frac{A_0^2 (0 + 1)}{2 + 0} \Rightarrow P = \frac{A_0^2}{2} \text{ Ans}$$



$$x[n] = u[n] \Rightarrow A_0 = 1$$

$$P = \frac{1}{2} \text{ W}$$

Discrete-Time Convolution

$$x[n] \rightarrow \text{LTI} \rightarrow y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k] \cdot h[n-k]$$

$n=k \downarrow$ $n=k \downarrow$
 $x[k]$ $h[k]$
 $\downarrow$ T.R. $\downarrow$ T.S. $\rightarrow$ $h[-k+n] = h[n-k]$

Example: Calculate $y[n]$.

$$u[n] \rightarrow \text{LTI} \rightarrow y[n] = u[n] * u[n] = \sum_{k=-\infty}^{+\infty} u[k] \cdot u[n-k]$$

Solution:

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Discrete Time Convolution (Tabular Method)

2. $x_1[n] = \{-1, 2, 0, 1\}$ $x_2[n] = \{2, 1, 0, -1\}$

$y[n] = x_1[n] * x_2[n] = ?$

	$x_2[n]$	-1	2	0	1
$x_1[n]$	-1	1	-2	0	-1
	2	-2	4	0	2
	0	0	0	0	0
	-1	1	-2	0	1

$y[n] = \{1, -2, 4, 0, 2\}$

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Discrete Time Convolution (Tabular Method)

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Static & Dynamic Discrete Time Systems

- 1) Static and Dynamic Systems
- 2) Causal and Non-Causal Systems
- 3) Time-Invariant and Time-Variant Systems
- 4) Linear and Non-Linear Systems
- 5) Stable and Unstable Systems

Static Systems: $y[n]$ depends only on the present $x[n]$

Dynamic Systems: $y[n]$ depends on past or future values of $x[n]$ at any instant of time

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10:16

Static & Dynamic Discrete Time Systems

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Causal & Non-Causal Discrete Time Systems

Causal Systems: $y[n]$ is independent of future values of $x[n]$

Non-Causal Systems: $y[n]$ depends on future values of $x[n]$ at any instant of time

1. $y[n] = x[n] + x[n+1]$ (Causal) $y[n] = x[n] + x[n-1]$ (Non-Causal)

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13:19

Causal and Non-Causal Discrete Time Systems

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TIV & TV Discrete Time Systems

For a system $y[n] = x[n]$ (TIV) $y[n] = x[n-1]$ (TV)

Homework

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8:47

Time-Invariant & Time-Variant Discrete Time Systems

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Linear & Non Linear Discrete Time Systems

System which follows the PRINCIPLE OF SUPERPOSITION is known as linear system

- Properties
- 1) System linearity is independent of time scaling
- 2) System linearity is independent of coefficient used in system relationship
- 3) If any system relationship exists other than linearity and it is possible in the system relationship then the system will be non-linear
- 4) If system is a combination of time shifted version of input, then the system will be linear
- 5) Integral and Differential operators are linear operators
- 6) Both and their combination are linear operators
- 7) In Real, Imaginary and Complex operators are nonlinear operators

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Linear and Non-Linear Discrete Time Systems

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Stable & Unstable

Stable & Unstable Discrete

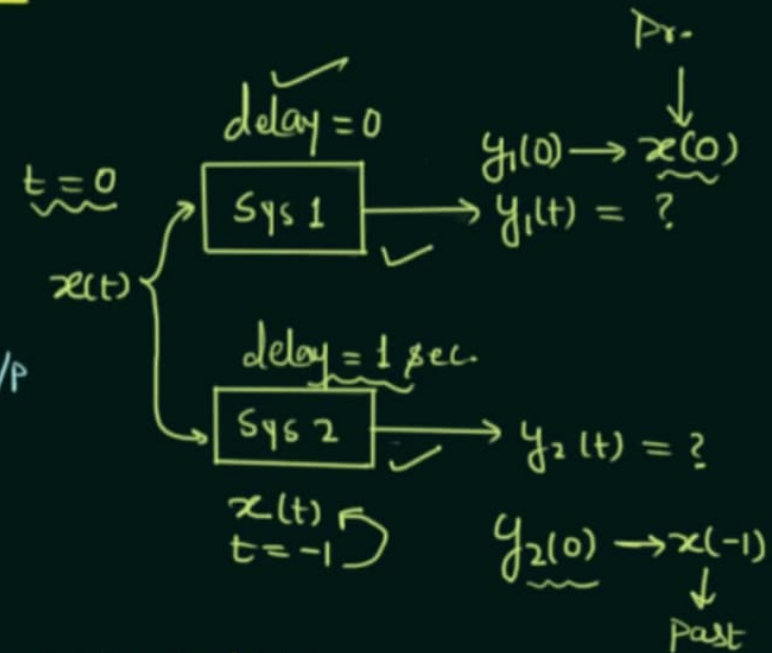
Causal and Non-Causal Systems

Causal System: o/p of system is independent of future values of i/p

ex: - i) $\check{y}(t) = \check{x}(t)$ C

ii) $\check{y}(t) = \underbrace{\check{x}(t)}_{\text{pr.}} + \underbrace{\check{x}(t-1)}_{\text{past}}$ C

C.S. $\rightarrow$ o/p $\rightarrow$ P i/p +
o/p $\rightarrow$ past i/p



Non-Causal System: o/p of system depends on future values of i/p at any instant of time

ex: - i) $\check{y}(t) = \underbrace{\check{x}(t+2)}_{\text{fut.}}$ N.C. + A.C.

ii) $\check{y}(t) = \underbrace{\check{x}(t)}_{\text{pr.}} + \underbrace{\check{x}(t-1)}_{\text{past}} + \underbrace{\check{x}(t+1)}_{\text{fut.}}$ N.C.

Anti-Causal System:

o/p $\rightarrow$ only on fut values of i/p

Causal and Non-Causal Systems (Solved Problems)

$$x(t) \rightarrow \downarrow \text{sys} \rightarrow y(t) = x(3t)$$

T.S.
↓

$$1) \quad \underline{y(t)} = \underline{x(3t)}$$

$$t=0$$

$$y(0) = x(0)$$

$$\underline{t=1}$$

$$\underline{y(1)} = \underline{x(3)}$$

fut. i/p

N.C.

$$2) \quad \underline{y(t)} = \begin{cases} x(3t) & t < 0 \\ \underline{x(t-1)} & t \geq 0 \end{cases}$$

$$t < 0$$

$$y(t) = x(3t)$$

$$\underline{t = -1}$$

$$\underline{y(-1)} = \underline{x(-3)} \quad \checkmark \subset$$

past i/p

$$t \geq 0$$

$$t=0$$

$$\underline{y(0)} = \underline{x(-1)} \quad \checkmark \subset$$

$$t=1$$

$$\underline{y(1)} = \underline{x(0)} \quad \subset$$

$\epsilon/2$ $\checkmark h = 2/\epsilon$

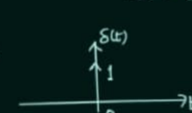
$$\delta(t) = \lim_{\epsilon \rightarrow 0} x(t) = \begin{cases} \infty, & t=0 \\ 0, & t \neq 0 \end{cases}$$

duration $\rightarrow 0$
 mag. $\rightarrow \infty$

1) $\int_{-\infty}^{\infty} \delta(t) dt = 1$

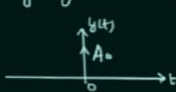
$\int_{-\infty}^{\infty} \delta(t) dt = \int_{-\infty}^{\infty} \lim_{\epsilon \rightarrow 0} x(t) dt = \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} x(t) dt$

Area of $\delta(t) = 1$



$y(t) = A_0 \delta(t)$
 $\int_{-\infty}^{\infty} y(t) dt = \int_{-\infty}^{\infty} A_0 \delta(t) dt = A_0 \int_{-\infty}^{\infty} \delta(t) dt = A_0 \cdot 1$

Area of $y(t) = A_0 = \text{weight of impulse}$



3) $\int \delta(t) dt = u(t)$

$\int A_0 \delta(t) dt = A_0 u(t)$
 weight $\uparrow$ $\uparrow$ step

Description

Properties of Impulse Signal (Part 1)

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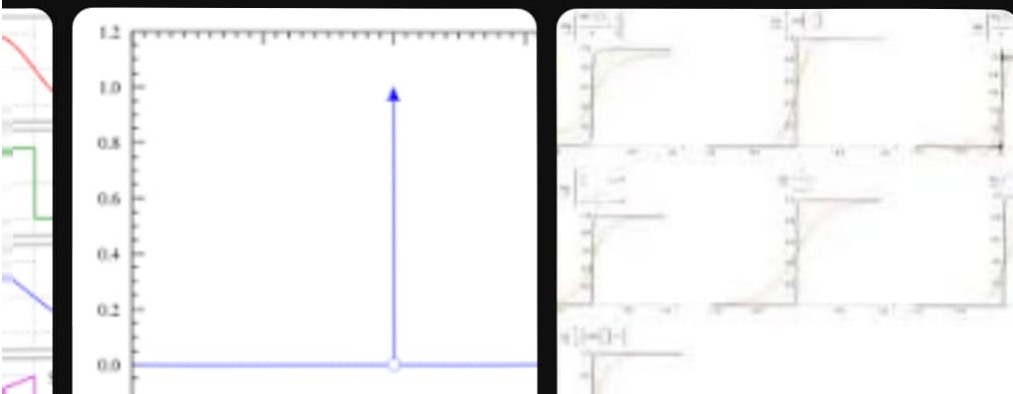
Signal and System: Properties of Impulse Signal (Part 1)

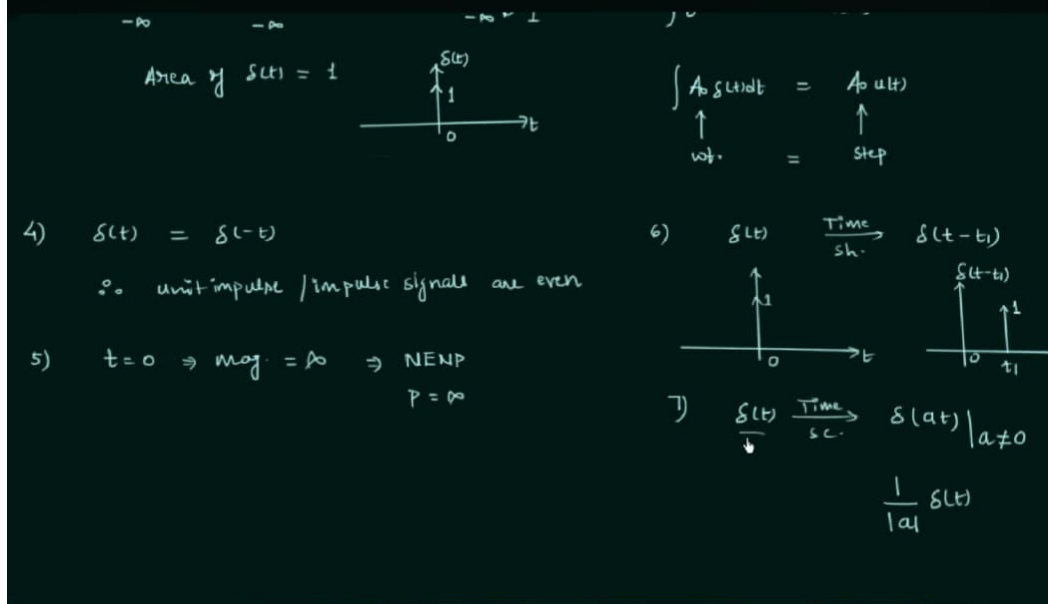
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2. Area of unit impulse signal.
3. Weight or strength of an impulse.
4. Impulse signal as an even signal ...more

Key concepts ⓘ

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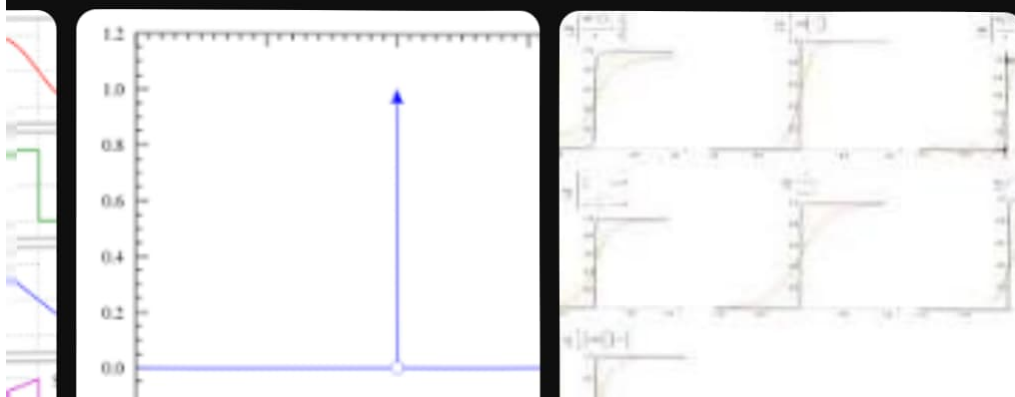
Signal and System: Properties of Impulse Signal (Part 1)

Topics Discussed:

1. Revision of unit impulse signal.
2. Area of unit impulse signal.
3. Weight or strength of an impulse.
4. Impulse signal as an even signal ...more

Key concepts ⓘ

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$$10) \int_{-\infty}^{\infty} x(t) \cdot \frac{d^n}{dt^n} \delta(t-t_1) dt = (-1)^n \frac{d^n x(t)}{dt^n} \Big|_{t=t_1}$$

$$\approx x(t) \Big|_{t=t_1} = 0 \text{ or finite}$$

$$\begin{aligned} \text{Ques:- } I &= \int_{-\infty}^{\infty} \cos \pi t \cdot \frac{d^2}{dt^2} \delta(t-1) dt \\ &= (-1)^2 \cdot \frac{d^2}{dt^2} (\cos \pi t) \Big|_{t=1} \\ &= 1 \cdot (-\pi^2 \cos \pi t) \Big|_{t=1} \\ &= -\pi^2 \times (-1) = \pi^2 \text{ Ans} \end{aligned}$$

Properties of Impulse Signal (Part 2)

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Magnitude Distortion in LTI system

Magnitude Distortion: If the system provides unequal amount of amplification to different frequency components available in input signal then system is having magnitude distortion

$$\begin{aligned} x(t) &\longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = \underbrace{A_1}_{\neq} \sin \omega_1 t + \underbrace{A_2}_{\neq} \sin \omega_2 t \\ &= \sin \omega_1 t + \sin \omega_2 t \quad \omega_1 \neq \omega_2 \end{aligned}$$

Magnitude Distortion in LTI Systems

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Phase Distortion

Phase Distortion: If the system provides unequal amount of time delays to different frequency components available in input signal then system is having phase distortion

$$\begin{aligned} x(t) &\longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = \sin[\omega_1(t-t_1)] + \sin[\omega_2(t-t_2)] \\ &= \sin \omega_1 t + \sin \omega_2 t \quad \omega_1 \neq \omega_2 \end{aligned}$$

136 / Signal & System

4:48



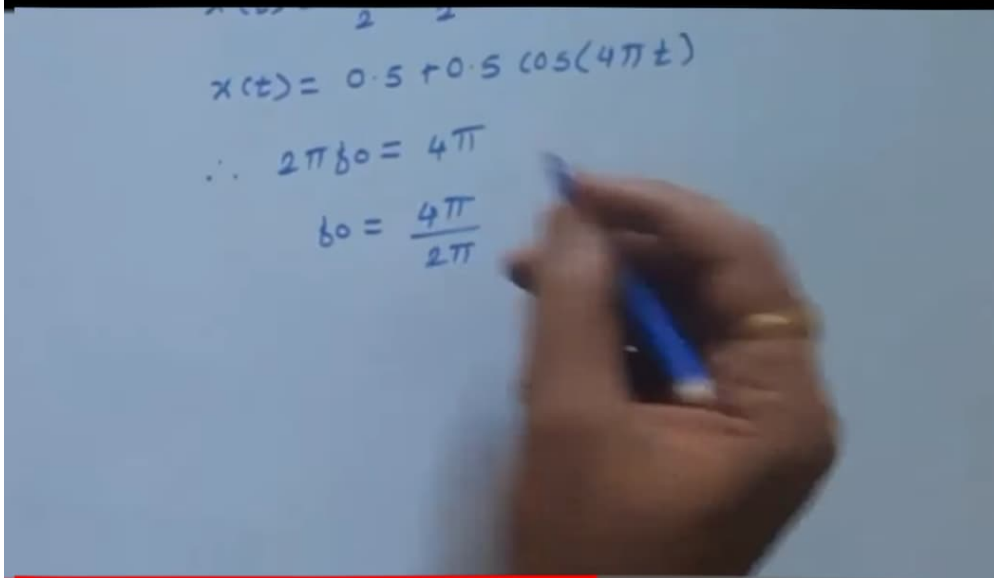
Phase Distortion in LTI Systems



Next: Phase Distortion in LTI Systems

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$$\begin{aligned}x(t) &= \cos^2(2\pi t) \\x(t) &= \frac{1 + \cos(2 \times (2\pi t))}{2} \\x(t) &= \frac{1}{2} + \frac{1}{2} \cos(4\pi t) \\x(t) &= 0.5 + 0.5 \cos(4\pi t) \\\therefore 2\pi f_0 &= 4\pi \\f_0 &= 1\end{aligned}$$

Description

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$$\begin{aligned}\therefore 2\pi f_0 &= 4\pi \\ f_0 &= \frac{4\pi}{2\pi} \\ f_0 &= 2 \text{ Hz} \\ \text{fundamental period } T_0 &\text{ is} \\ T_0 &= \frac{1}{f_0} = \frac{1}{2} = 0.5 \text{ sec.} \\ \therefore T_0 &= 0.5 \text{ sec} \\ \therefore \text{Given signal } x(t) &\text{ is periodic} \\ &\text{with period } 0.5\end{aligned}$$

Description

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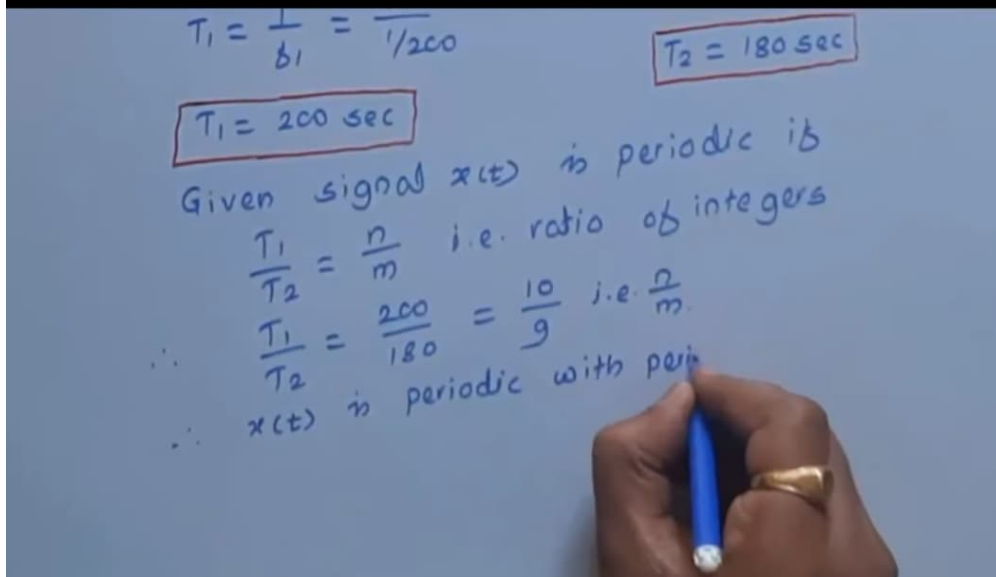
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$$1) x(t) = 5 \cos(200\pi t)$$

$\Rightarrow$ Comparing given signal with standard cosine signal

$$x(t) = A \cos(2\pi f_0 t)$$

then we will get

$$2\pi f_0 = 200\pi$$

$$f_0 = \frac{200\pi}{2\pi}$$

Part1: Examples of Periodic & Non-periodic Signals Part1 | Signals & ...

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Hello Sir, thank you so much for your video. Please I would like to kindly draw your attention to the last sol...

* For given signal, determine whether it is periodic or Non-periodic & if periodic find its period.

$$1) x(t) = 3e^{-j12t}$$

$$2) x(n) = \sin\left(\frac{100}{9}\pi n\right) + \cos\left(\frac{200}{7}\pi n\right)$$

$$3) x(n) = \sin(145\pi n) + \cos(170\pi n)$$

$$4) x(n) = (-1)^n$$

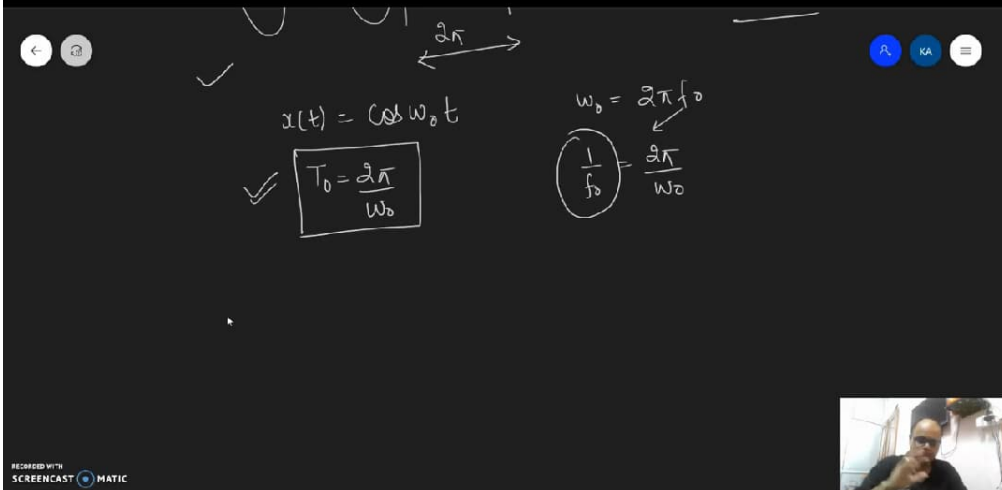
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Part2: Examples of Periodic and Non-periodic Signals | Signals & Systems

techAj · 28K views · 3 years ago





Periodic and Aperiodic Signals [Continuous & Discrete Time Periodic ...

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KK Awasthi EduHub 18.5K



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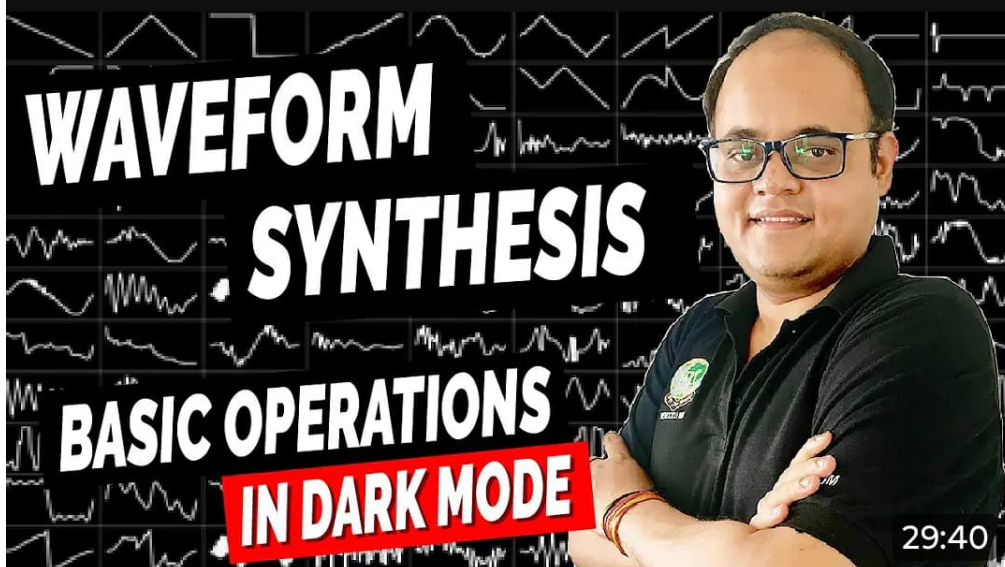


Thanks

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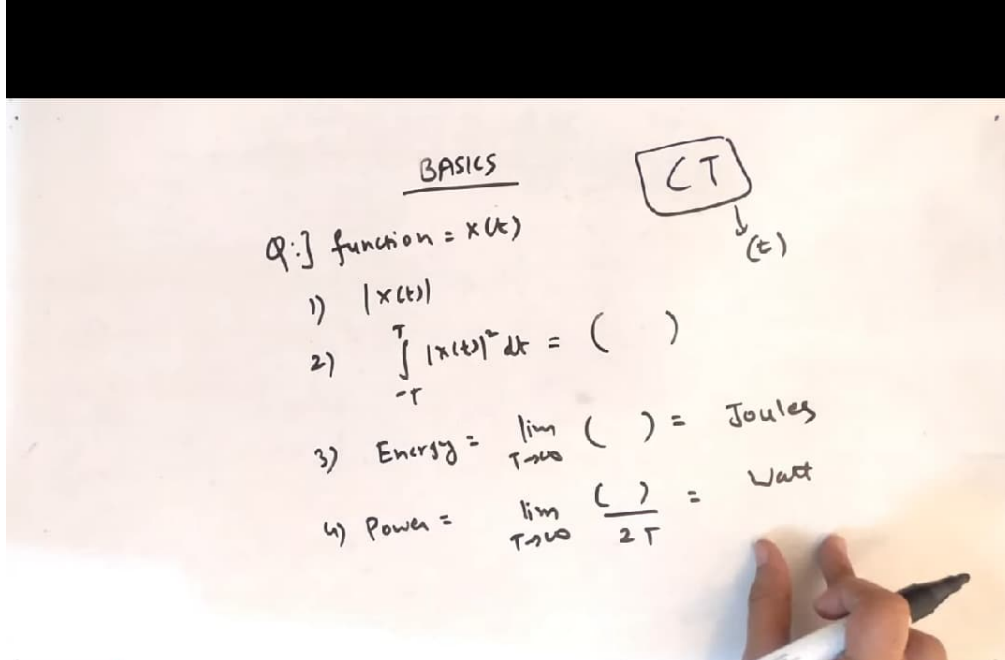
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Waveform Synthesis-Basic Operations
[Time Shifting, Time Scaling, Time Reversa...



KK Awasthi EduHub · 23K views · 3 years ago



Energy and Power Signals | Solved Problems / Examples

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SHRENIK JAIN - STUDY SIMPLIFIED

ENERGY & POWER
SIGNAL

BEST
VIDEO

DISCRETE TIME SIGNAL



1 LAKH + VIEWS



19:30



Energy and Power Signal (DT) - Examples, Calculations, Equation, Formulae, Tricks

Shrenik Jain · 1.3 lakh views · 6 years ago

Energy and Power Signals

$x(t) = e^{j\omega t}$

~~$|e^{j\omega t}|$~~

$$|x(t)|^2 = A^2$$

$$\lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \underline{\underline{A^2}}$$

$$\cos \theta + j \sin \theta$$

$$\sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$

$$x(t) = 5 e^{j \frac{2\pi}{11} t}$$

25 $\leftarrow e^{j\theta}$

Energy and Power Signals

$$x(t) = Ae^{j\omega t}$$

$$|x(t)| = |A| |e^{j\omega t}|$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$|e^{j\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$$

$$|x(t)|^2 = A^2$$

$$\lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \underline{\underline{A^2}}$$

$$x(t) = 5 e^{-\frac{j2\pi}{11}t}$$

25 ← $e^{j\theta}$

Q: Find which of the following signals are power signals, neither energy nor power signals.

- a) $(\frac{1}{2})^n u(n)$ b) $e^{j[\frac{\pi}{3}n + \frac{\pi}{2}]}$ c) $\sin(\frac{\pi}{3}n)$
 d) $u(n) - u(n-6)$ e) $n u(n)$ (f) $x(n) - x(n-4)$

Solⁿ:

Energy signal

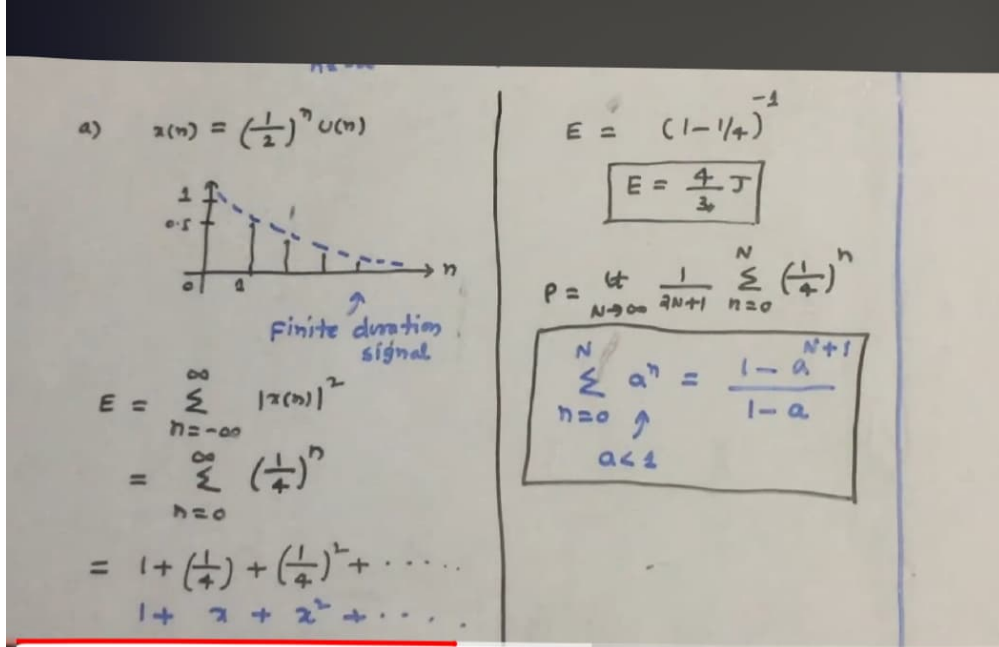
$$0 < E < \infty, \quad p = 0$$

$$\checkmark E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

power signal

$$0 < p < \infty; \quad E = \infty$$

$$\checkmark p = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$



Signals & Systems - Energy & Power of Discrete time signals - working exa...

26K views 3 yr ago ...more



Dr.P.Prasanna Murali krishna 25.8K

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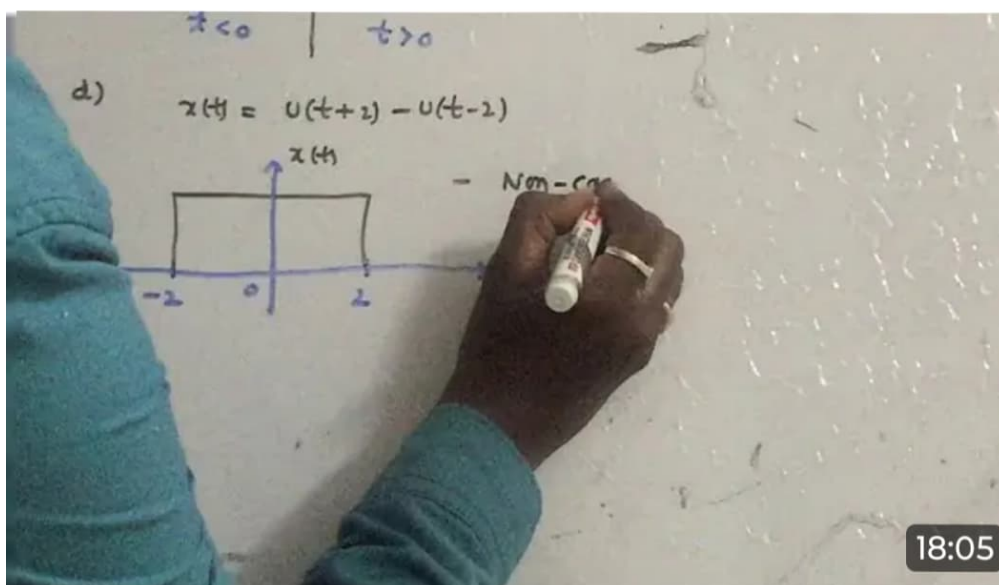


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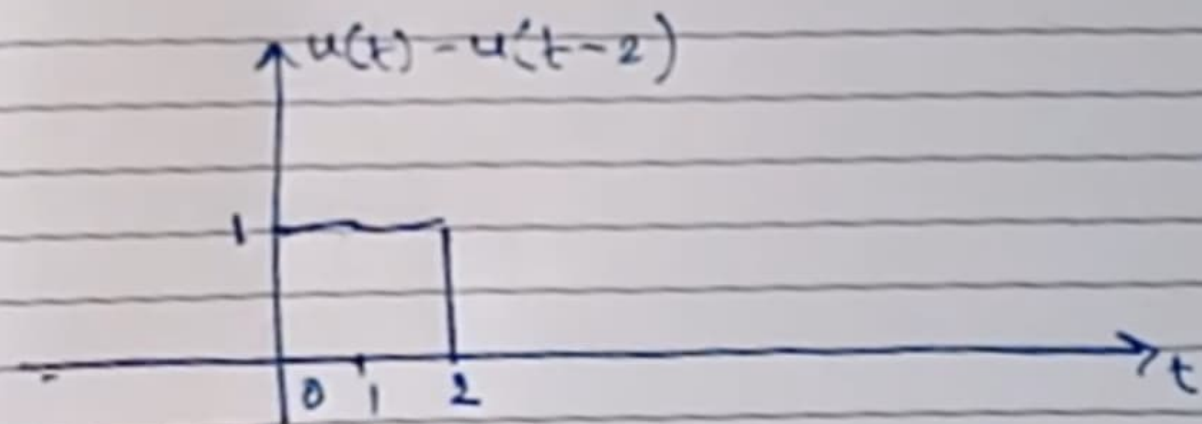
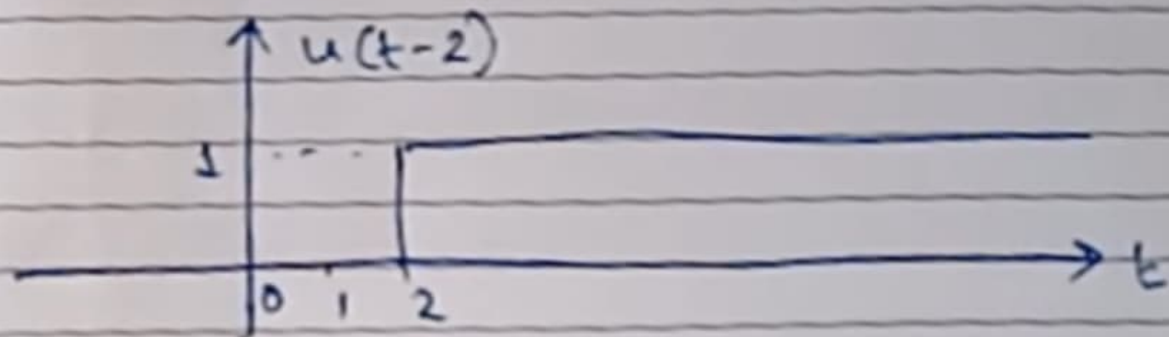
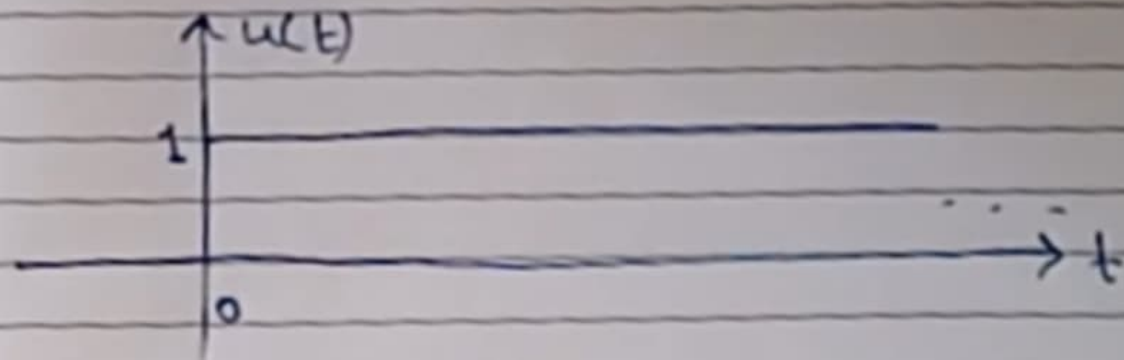
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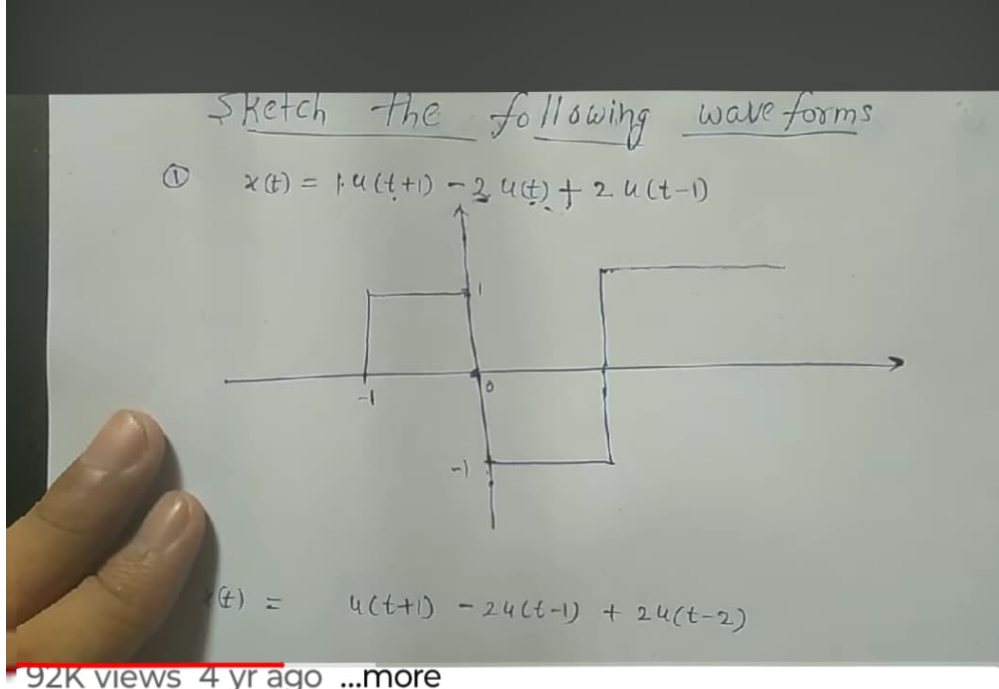


Signals & Systems - Causal & Non-causal signals - working examples

Dr.P.Prasanna Murali krishna · 13K views · 3 years ago

$$1) x(t) = u(t) - u(t-2)$$





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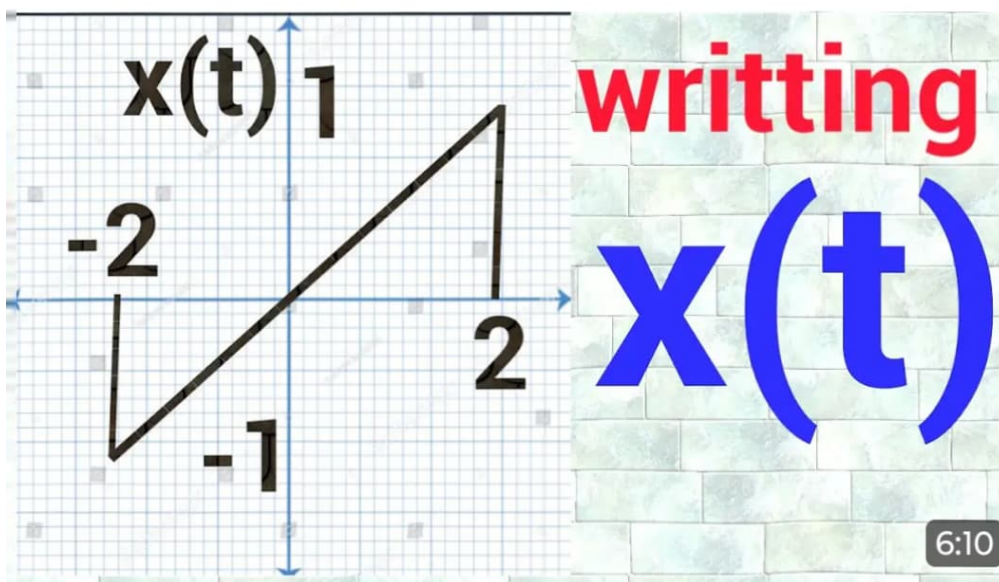


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Writting signal from waveform | signals and systems

easy study plus · 3.9K views · 4 years ago

Sketch the following signals

$x[n] = \{ 1, -1, 2, 0, 1, 1, 2 \}$
 $n=0$

Obtain :

i) $x[n]$	vi) $x[4-2n]$
ii) $x[-n]$	vii) $x[n] \cdot u[n-1]$
iii) $x[n-2]$	viii) $x[n-1] \cdot \delta[n-2]$
iv) $x[3-n]$	
v) $x[2n+1]$	



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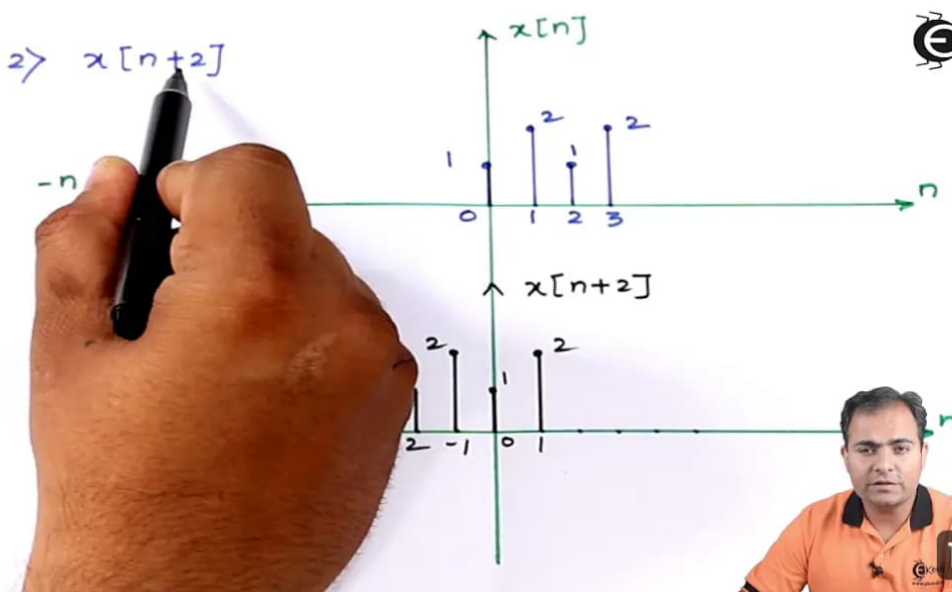
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2) $x[n+2]$

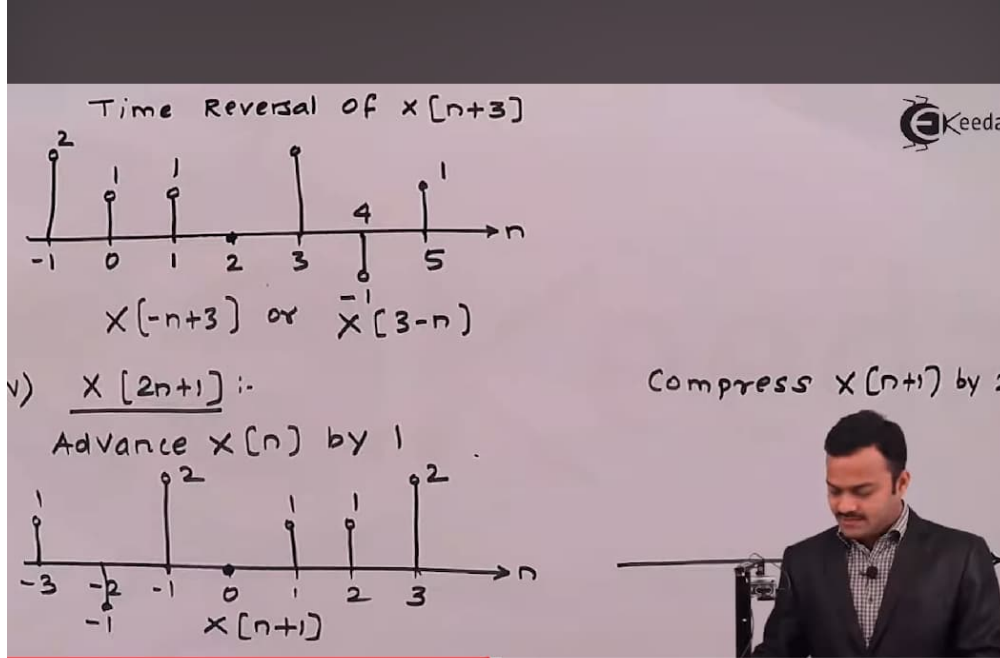



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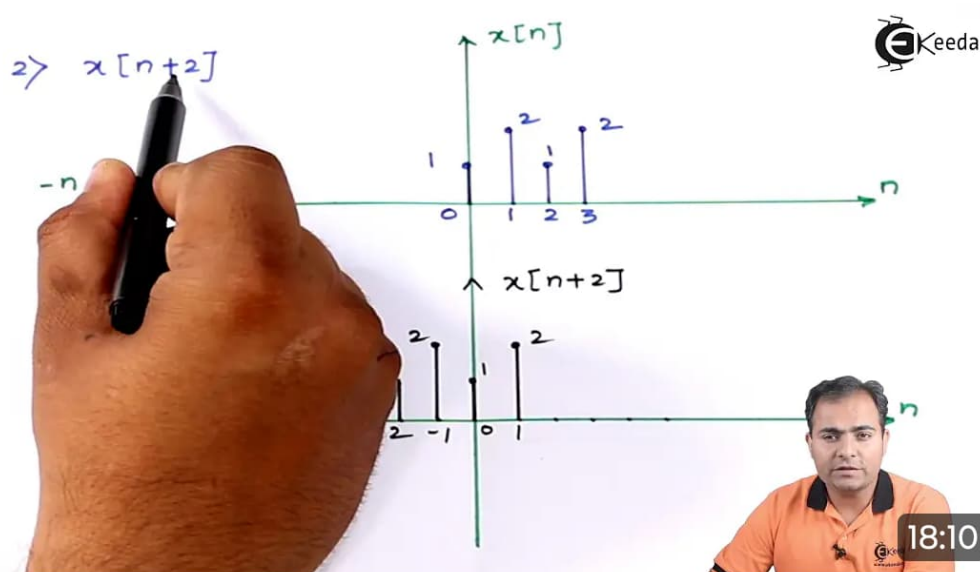


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4 Apr

Welcome to our channel! In this enlightening video, we delve into the intriguing realm of the unit parabolic function—a pivotal concept in the study of signals and systems. Whether you're a student striving for clarity or an enthusiast hungry for knowledge, this video is tailored to broaden your understanding ...more

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Convolution - CT Signals

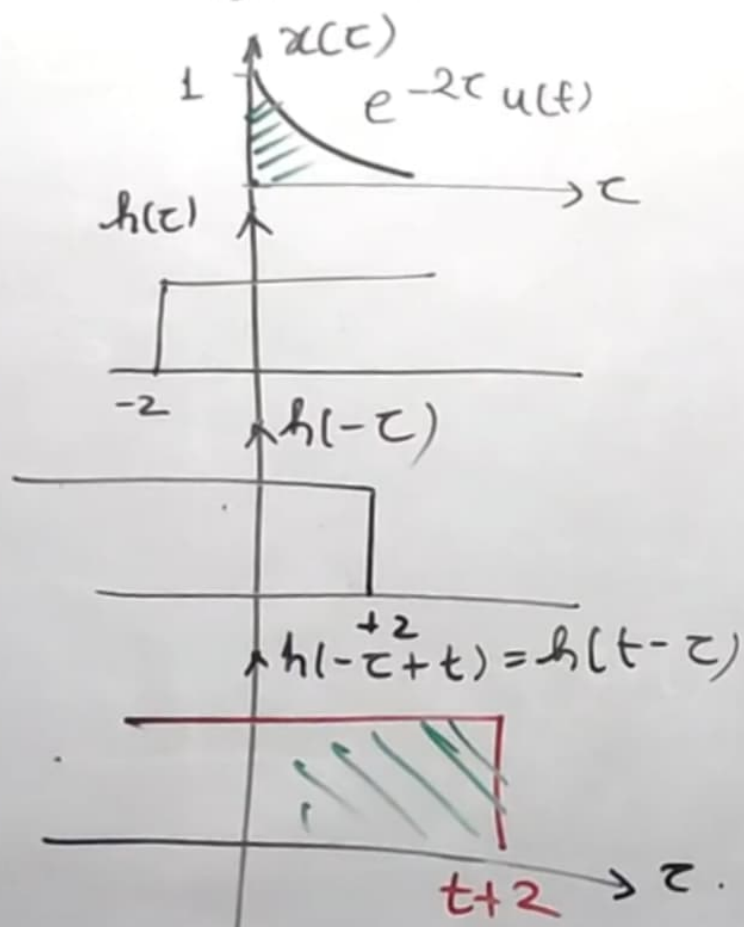
Exp * Step

$$x(t) = u(t-1)$$
$$h(t) = u(t+2)$$

Case 1

$$x(t) = e^{-2t} u(t)$$

$$h(t) = u(t+2)$$



$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$h(-\tau) \rightarrow h(-\tau+t)$$

$$-x+T \rightarrow \text{Right}$$

$$-x-T \rightarrow \text{Left}$$

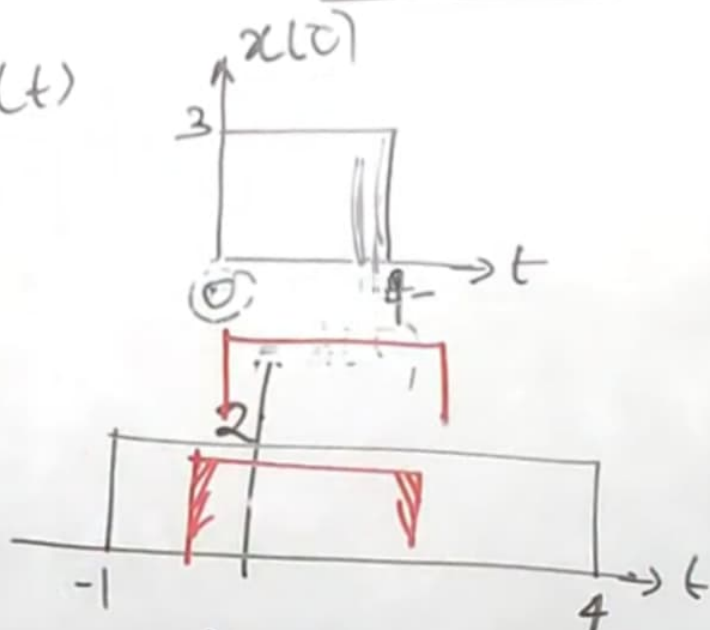
Convolution - CT Signals

Case 1

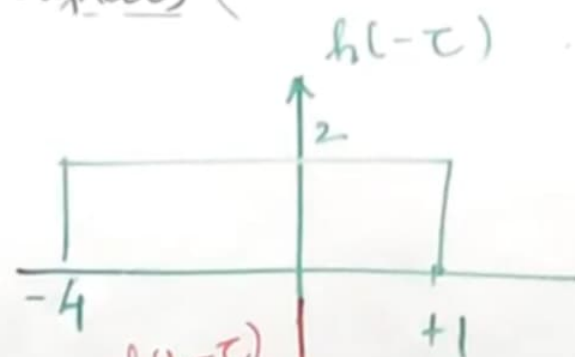
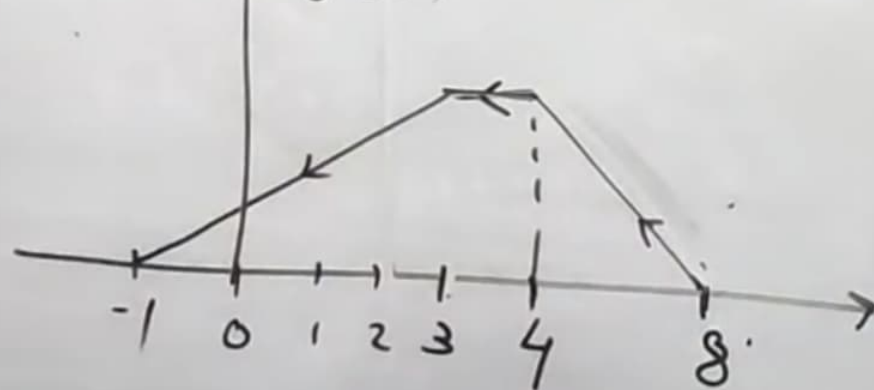
$$t - 4 = 4$$

$$t = 8$$

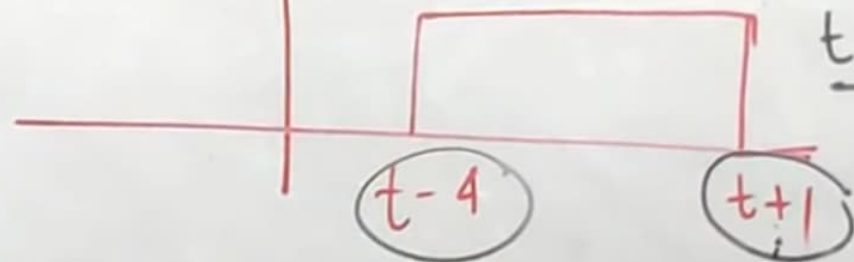
$x(t)$



$y(t)$



$h(t-\tau)$



$$t+1 = 0$$

$$t = -1$$

$h(-\tau+t)$

$$t > 0$$

$$t - 4 = 0$$

$$t = 4$$

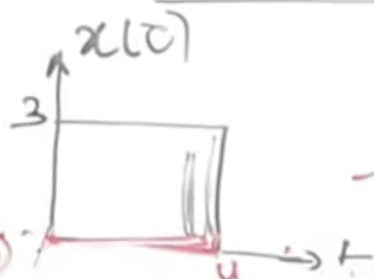
$$t+1 = 4$$

$$t = 3$$

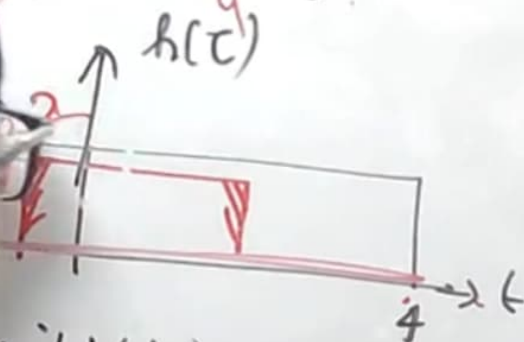
Convolution - CT Signals

Case 1

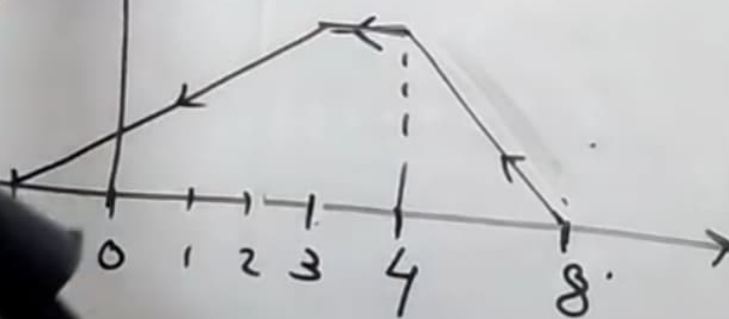
$x(t)$



$$-1 + 4 = 3$$

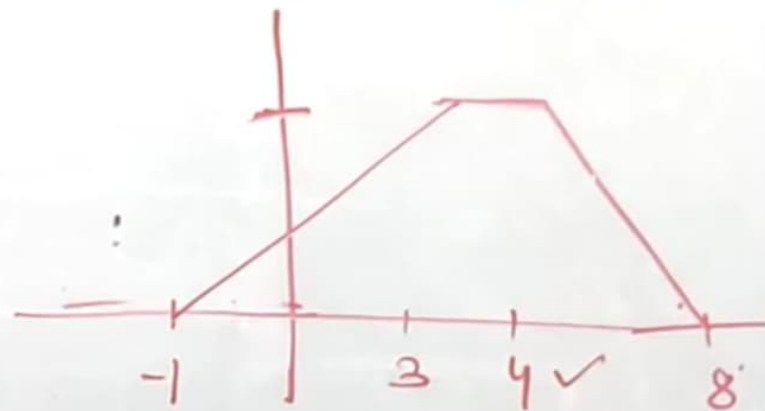


$y(t)$



$$x_e + h_e = 0 - 1 = -1$$

$$x_h + h_h = 8$$



$$A_1 \times A_2 \times 4$$

=

$$y(t) = s(t) = \int_{-\infty}^t h(\tau) u(t-\tau) d\tau$$

$$s(t) = \int_{-\infty}^t h(\tau) d\tau$$

$$s(t) = \int_{-\infty}^t h(t) dt$$

Step Response = $\int$ Impulse Response

Impulse Response = $\frac{d}{dt}$ (Step Response)



$$S(t) = \int_{-\infty}^t h(t) dt = \int_{-\infty}^t t u(t) dt = \int_0^t t dt =$$

$$b) \quad S(t) - S(t-1)$$



Navigation and control icons:

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- Fullscreen icon

← ↺ given →

$$\text{Response} = \frac{d}{dt} (\text{Response})$$

KA

Q: a) $h(t) = t u(t)$ Step Response?

$$S(t) = \int_{-\infty}^t h(t) dt = \int_{-\infty}^t t u(t) dt = \int_0^t t dt = \frac{t^2}{2} u(t)$$

b) $S(t)$



$$= u(t) - u(t-1) \quad f_2$$

c)

$$h(t) = u(t+1) - u(t-1)$$

$$s(t) = \int_{-\infty}^t u(\tau+1) d\tau - \int_{-\infty}^t u(\tau-1) d\tau$$

$$= \int_{-1}^t 1 d\tau - \int_1^t 1 d\tau = \left. \tau \right|_{-1}^t - \left. \tau \right|_1^t$$

$$= t - 1 - (t - 1)$$

$$= \underline{\underline{0}} \quad \checkmark$$



Trigonometric Fourier Series

$x(t) =$ dc / avg. value of $x(t)$ + cosine terms + sine terms

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

Ex:- $x(t) = \underbrace{4}_{a_0} + \underbrace{3 \cos 2\omega_0 t}_{2^{\text{nd}} \text{ harmonic}} + \underbrace{5 \sin 2\omega_0 t}_{2^{\text{nd}} \text{ harmonic}} + \underbrace{4 \cos 3\omega_0 t}_{3^{\text{rd}} \text{ harmonic}} + \underbrace{2 \sin 3\omega_0 t}_{3^{\text{rd}} \text{ harmonic}} + \dots$

Fourier Coefft:

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin n\omega_0 t dt$$

$$a_2 = 3$$

$$b_2 = 5$$

$$a_2 < b_2 \Rightarrow \text{more sine}$$

$$a_3 = 4$$

$$b_3 = 2$$

$$a_3 > b_3 \Rightarrow \text{more cosine}$$

Trigonometric Fourier Series

4.5 lakh views 6 yr ago #FourierSeries ...more



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4.5K

• 120



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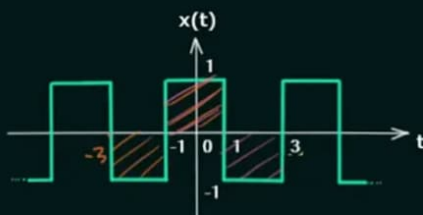


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"Symmetrical about t-axis" it means that the positive and negative area is equal in magnitude.

Trigonometric Fourier Series (Example 1)



$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

$$x(t) = \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos n\omega_0 t dt$$

$$a_n = \frac{1}{2} \int_{-1}^3 x(t) \cos \frac{n\pi}{2} t dt$$

$$T_0 = 4 \text{ sec}$$

$$\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{2} \text{ rad/sec}$$

$$\omega_0 = \frac{\pi}{2} \text{ rad/sec}$$

144 / Signal & System

26:39



Trigonometric Fourier Series (Example 1)

Next: Trigonometric Fourier Series (Example 1)

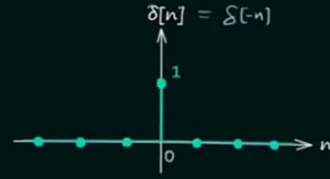
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DRYBAR

Unit Impulse Sequence

Properties:

1. $\delta[n]$ is an Even Signal.
2. $\delta[n]$ is an Energy Signal.
3. $\delta[kn] = \delta[n]$
5. $x[n] * \delta[n-n_1] = x[n-n_1]$
6. $\sum_{k=-\infty}^{\infty} \delta[k] = u[n]$
7. $u[n] - u[n-1] = \delta[n]$



$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

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Unit Impulse Sequence

30K views

